

Gene, Transcript, Protein, and Perturbation Identifier Systems

A first-principles monograph for the MANTRA perturbation-response data stack

Revised draft - July 2026

Abstract

Biological data commonly place gene symbols, gene-record accessions, transcript accessions, protein accessions, guide identities, assay accessions, and model-coordinate keys beside one another. These strings are related, but they identify different kinds of objects. This monograph develops the relevant molecular biology from first principles, explains how genome annotations and biological databases are constructed, gives a detailed account of single- and dual-guide CRISPR perturbation designs, and then specifies the identifier architecture used by the MANTRA perturbation-response pipeline. The central design principle is that biological identity is a typed, release-aware record with several orthogonal projections rather than one universal name.

Contents

1 The Biological Identity Problem.....	7
1.1 Why a gene-identification system becomes difficult in practice.....	7
1.2 Four categories of objects.....	8
1.3 A typed model of identity.....	8
1.4 Orthogonality as a database-design concept.....	9
1.5 Scope of this monograph.....	9
2 Biological Foundations.....	10
2.1 DNA as a physical molecule and an information-bearing polymer.....	10
2.2 Genome, assembly, chromosome, contig, and coordinate.....	10
2.3 The historical and modern meanings of a gene.....	11
2.4 Locus boundaries, promoters, enhancers, and regulatory regions.....	11
2.5 Transcription: synthesis of RNA from a DNA template.....	11
2.6 RNA processing and alternative splicing.....	12
2.6.1 Exon skipping.....	13
2.6.2 Alternative splice sites.....	13
2.6.3 Intron retention.....	13
2.6.4 Alternative transcript starts and ends.....	13
2.7 Isoforms: a precise distinction among transcript, protein, and modification states.....	13
2.8 Translation: from coding RNA to protein.....	14
2.9 Coding and noncoding as annotation properties.....	14
2.10 Major classes of noncoding RNA.....	14
2.11 Long noncoding RNAs in depth.....	15
2.11.1 lincRNA, lncRNA, and related labels.....	16
2.11.2 Expression and evidence.....	16
2.12 cDNA: complementary DNA as an experimental representation of RNA.....	16
2.12.1 cDNA, genomic DNA, and transcript models.....	17
2.13 Pseudogenes in depth.....	17
2.13.1 Duplicated or unprocessed pseudogenes.....	18
2.13.2 Processed pseudogenes.....	18
2.13.3 Unitary pseudogenes.....	18
2.13.4 Transcribed and translated pseudogene evidence.....	19
2.13.5 Why pseudogene symbols often resemble parent-gene symbols.....	19
2.13.6 Pseudogenes in expression data.....	19
2.14 Paralogs, orthologs, and gene families.....	20
2.15 A compact biological object graph.....	20

3 Genome Annotation.....	21
3.1 Assembly and annotation solve different problems.....	21
3.2 What an annotation team actually constructs.....	21
3.3 Sources of annotation evidence.....	22
3.3.1 Short-read RNA sequencing.....	22
3.3.2 Long-read RNA sequencing.....	22
3.3.3 Full-length cDNA and expressed-sequence tags.....	22
3.3.4 Protein evidence.....	22
3.3.5 Comparative genomics.....	23
3.3.6 Manual curation and literature.....	23
3.4 Gene models and transcript models.....	23
3.5 Why transcript annotations change.....	23
3.6 Canonical, representative, principal, and MANE transcripts.....	24
3.7 GTF and GFF as annotation exchange formats.....	24
3.8 Gene and transcript biotypes.....	25
3.9 Genome assembly patches and alternate loci.....	25
3.10 Coordinate transformations and lift-over.....	25
3.11 Release pinning as scientific state management.....	25
4 The Identifier Systems.....	27
4.1 The central claim: these nomenclatures are orthogonal.....	27
4.2 A question-to-resource matrix.....	28
4.3 HGNC: human gene nomenclature.....	28
4.3.1 Approved symbols, previous symbols, and aliases.....	29
4.3.2 HGNC locus types.....	29
4.3.3 Example: ALG1L and ALG1L1P.....	29
4.4 Ensembl: stable identifiers for genome features.....	29
4.4.1 Ensembl gene, transcript, and protein relations.....	30
4.4.2 Species context.....	30
4.5 GENCODE: a release-specific human and mouse gene annotation.....	30
4.5.1 Comprehensive and basic annotations.....	31
4.5.2 GENCODE tags and transcript support.....	31
4.6 NCBI Gene: stable gene records in the NCBI ecosystem.....	31
4.7 RefSeq: accessioned reference sequences.....	31
4.8 UniProt: protein knowledge and sequence identity.....	32
4.9 ENCODE: identifiers for functional-genomics data.....	32
4.10 Genome Reference Consortium and assembly identifiers.....	33
4.11 MANE and CCDS as convergence layers.....	33

4.12 A formal crosswalk model.....	33
4.13 Identifier syntax reference.....	34
4.14 Orthogonality in one concrete row.....	34
5 Perturbation Experiments and Guide Architecture.....	35
5.1 Perturbation as an experimental object.....	35
5.2 CRISPR-Cas targeting from first principles.....	35
5.3 CRISPR nuclease knockout.....	35
5.4 CRISPR interference.....	36
5.5 CRISPR activation.....	36
5.6 The semantic layers of a guide record.....	36
5.7 Single-guide library elements.....	37
5.8 Dual-guide elements targeting one gene.....	37
5.8.1 Why two guides can improve knockdown.....	38
5.8.2 Guide A and guide B as positional identities.....	38
5.9 Paired guides targeting different genes.....	38
5.10 Paired guides for genomic deletions.....	38
5.11 The sgID_AB field in the K562 atlas.....	39
5.12 Direct guide capture.....	39
5.13 Linked barcodes and direct capture.....	39
5.14 Lentiviral delivery and recombination context.....	40
5.15 Multiplicity of infection, singlets, and multiple guide assignments.....	40
5.16 Target identity versus response identity.....	40
5.17 Controls.....	40
5.18 Perturbation identity hierarchy.....	41
6 Single-Cell Perturbation Data.....	42
6.1 What Perturb-seq measures.....	42
6.2 From cellular RNA to sequencing reads.....	42
6.3 Cell barcodes.....	43
6.4 Unique molecular identifiers.....	43
6.5 Read alignment and gene assignment.....	43
6.6 Counts, normalization, and transformed expression.....	43
6.7 Pseudobulk aggregation.....	44
6.8 AnnData and H5AD.....	44
6.9 The two dimensions of an expression matrix.....	44
6.10 The raw K562 atlas as an annotated matrix.....	45
6.11 Direct guide capture as a second feature space.....	45
6.12 Controls and matched-control baselines.....	45

6.13 Data provenance from molecule to model.....	45
7 MANTRA Identifier Architecture.....	46
7.1 Architectural goal.....	46
7.2 Canonical object sets and notation.....	46
7.3 Raw cell-level fields.....	47
7.4 Perturbation split contract.....	48
7.5 Why the perturbation label remains historical.....	48
7.6 Expression-gene metadata.....	48
7.7 Construction of the 5,000-gene response axis.....	49
7.8 TBCE: duplicate display symbol, distinct response coordinates.....	49
7.9 The two axes as distinct types.....	50
7.10 Identifier flow.....	50
7.10.1 Perturbation branch.....	50
7.10.2 Response branch.....	50
7.11 Required resolved identity record.....	50
7.12 Resolution precedence.....	51
7.13 Using Ensembl IDs for locus and matrix reconciliation.....	51
7.14 Transcript interpretation.....	52
7.15 Protein-derived prior coverage.....	52
7.16 Coordinate-derived prior coverage.....	52
7.17 Project-local identifiers.....	52
7.18 Source artifact identity.....	53
8 Worked Identity Case Studies.....	54
8.1 Case study 1: ALG1L across experimental and current contexts.....	54
8.2 Case study 2: TBCE and response-axis uniqueness.....	55
8.3 Case study 3: TP53 across gene, transcript, sequence, and protein namespaces.....	55
8.4 Case study 4: a long noncoding RNA locus.....	55
8.5 Case study 5: a processed pseudogene.....	56
8.6 Case study 6: one dual-guide perturbation record.....	56
8.7 Case study 7: one cell from physical experiment to model tensor.....	56
8.8 Case study 8: ENCODE signal summarized to a gene prior.....	57
9 Implementation and Governance.....	58
9.1 The identity layer as a typed subsystem.....	58
9.2 Parsing identifier syntax.....	58
9.3 Resolution result as a first-class object.....	59
9.4 Crosswalk tables.....	59
9.5 Ordered axis contracts.....	60

9.6 Invariants for the K562 atlas contract.....	60
9.7 Build reports and manifests.....	60
9.8 Suggested identity schema.....	61
9.8.1 Core perturbation table.....	61
9.8.2 Raw experimental mapping table.....	61
9.8.3 Protein mapping table.....	62
9.8.4 Response-axis table.....	62
9.9 API design.....	62
9.10 Diagnostic summaries.....	63
9.11 Versioned policy.....	63
9.12 Comparison between resolver versions.....	63
9.13 Provenance hashes and content addressing.....	64
9.14 Human-readable documentation and machine-readable contracts.....	64
9.15 Canonical operational summary.....	64
10 Reference Appendices.....	66
Appendix A. Mental models.....	66
A.1 The biological production chain.....	66
A.2 The experimental observation chain.....	66
A.3 The database projection chain.....	66
Appendix B. Object-level reference table.....	66
Appendix C. Prefix and suffix reference.....	67
Appendix D. Guide-pair terminology.....	68
Appendix E. Glossary.....	68
Appendix F. Evidence inventory for the MANTRA implementation.....	71
Appendix G. Selected references and authoritative resources.....	72
Appendix H. Final canonical answer.....	74

1 The Biological Identity Problem

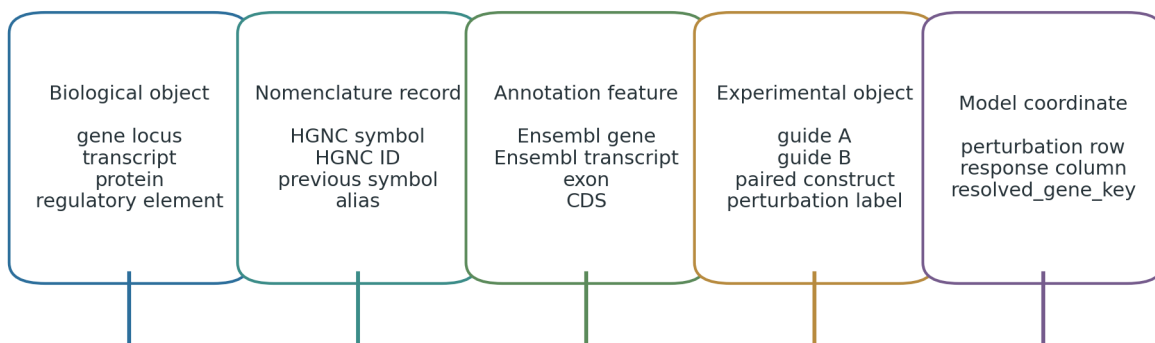
1.1 Why a gene-identification system becomes difficult in practice

A software system often begins with an apparently simple requirement: every perturbation and every measured gene should have a stable key. The requirement becomes difficult as soon as data are integrated across experimental generations and biological resources. A perturbation library may name a target with the gene symbol that was current when the library was designed. A contemporary nomenclature database may now use a different approved symbol. The raw single-cell file may also carry an Ensembl gene identifier and a transcript identifier. A protein-derived prior may require a UniProt accession. A chromatin-derived prior may originate from an ENCODE file interpreted against a GENCODE annotation on a particular genome assembly. The prediction tensor may finally use a project-local coordinate that was created to preserve duplicate display symbols.

All of these identifiers can describe one connected biological case. They still have different semantics. The guide-pair identifier names an experimental construct. The HGNC ID names a human gene-nomenclature record. The Ensembl transcript accession names one annotated transcript feature. The UniProt accession names a protein knowledgebase entry. The ENCODE accession names an assay or data file. The MANTRA resolved_gene_key names one model coordinate. A robust system preserves those identities as related fields instead of compressing them into a single string.

The practical source of confusion is that the word *gene* is used at several levels. It can mean a biological locus, a database record about that locus, a human-readable symbol, a row in a perturbation split, a column in an expression matrix, or the intended target of a guide construct. The surrounding context usually disambiguates the meaning in ordinary scientific prose. A computational pipeline has to make the disambiguation explicit.

Biological identity is a product of orthogonal object spaces



A complete identity record preserves each projection and the release-specific relations among them.

**Typed identity = namespace + object type + accession + version
+ release + assembly + provenance**

Biological identity is distributed across orthogonal object spaces.

1.2 Four categories of objects

The identity layer becomes easier to reason about when its objects are divided into four broad categories.

Category	Examples	What establishes identity
Biological objects	genomic locus, RNA transcript, protein molecule, regulatory element	biological structure and relation to other biological objects
Database records	HGNC record, Ensembl feature, RefSeq sequence, UniProt entry	namespace, accession, version, and database release
Experimental objects	perturbation label, guide A, guide B, paired cassette, sequenced cell	library design, physical construct, sample provenance, and assay metadata
Computational objects	H5AD row, H5AD column, split row, tensor coordinate, resolved_gene_key	an explicit data contract and ordered axis definition

These categories overlap through mappings. A guide construct targets a genomic region near the promoter of a gene. An Ensembl transcript belongs to an Ensembl gene within a particular annotation release. A UniProt protein entry can be cross-referenced to a gene record and one or more transcript products. A response coordinate can be reconciled to an Ensembl gene ID. The mappings provide the connections while each category retains its own object semantics.

1.3 A typed model of identity

A portable identifier can be represented as a typed tuple

$$I = (n, o, a, v, r, b, p),$$

where:

- n is the namespace, such as HGNC, Ensembl, RefSeq, UniProt, ENCODE, or MANTRA;
- o is the object type, such as gene record, transcript feature, protein entry, assay, guide pair, or model coordinate;
- a is the accession or local key;
- v is the feature or sequence version when the namespace uses one;
- r is the database release, snapshot, or exact source-file identity;
- b is the genome assembly when genomic coordinates are involved; and
- p is provenance describing the file, resolver, or procedure that asserted the identity.

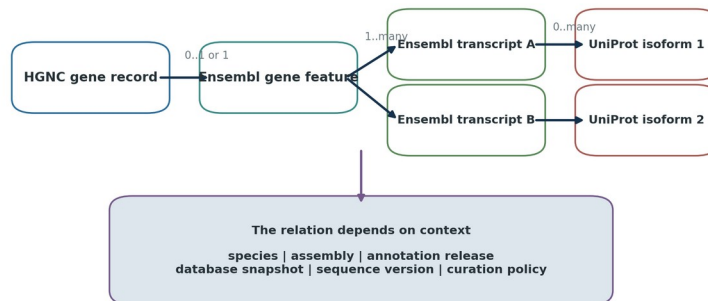
The tuple is intentionally broader than the visible identifier string. ENST00000611639.4 contains the Ensembl transcript accession and feature version, while a reproducible interpretation also records the Ensembl or GENCODE release and assembly. HGNC:33721 is a stable record identifier without a suffix version, so the snapshot or source-file hash supplies the time-specific context. ALG1L is a meaningful experimental label when its field is identified as the perturbation row inherited from the library. The surrounding typed record supplies its namespace, object type, release, and mapping status.

1.4 Orthogonality as a database-design concept

The biological resources discussed in this document are best understood as orthogonal services. In relational-database terms, they maintain different tables with different primary keys and different functional dependencies. HGNC is primarily a nomenclature authority. GENCODE is an annotation release. Ensembl is a genome database with stable feature identifiers. RefSeq is a collection of accessioned reference sequences. UniProt is a protein knowledgebase. ENCODE is an experimental data portal. The MANTRA split contract and response axis are project-local ordered coordinate systems.

The overlap among these resources is deliberate. A gene record can have cross-references into several namespaces because the resources describe complementary properties. The corresponding crosswalks are usually neither universal bijections nor timeless functions. A gene can have many transcripts. A transcript may produce a protein or may be noncoding. One gene can have several protein isoforms. A historical symbol can resolve to a current nomenclature record. A feature can be revised between annotation releases. A database record can be merged, split, withdrawn, or reclassified. The software representation should therefore treat cross-resource mappings as release-specific relations.

Cross-database mappings are release-specific relations rather than universal one-to-one functions



A reproducible join records both endpoints and the exact crosswalk that asserted their relationship.

Cross-resource mappings have explicit cardinality and release context.

Concept box - A symbol is a label, an accession is a record key, and a coordinate is an axis position.

These three ideas frequently appear in the same table. A symbol such as TP53 is optimized for human communication. An accession such as ENSG00000141510 identifies a database feature record. A model coordinate such as response-column 1,642 identifies a location in a tensor. A project can store all three because they answer separate questions.

1.5 Scope of this monograph

The next chapter develops the molecular biology needed to understand genes, transcripts, proteins, coding potential, cDNA, long noncoding RNAs, pseudogenes, and isoforms. Chapter 3 explains genome assembly and annotation. Chapter 4 presents the major identifier systems as orthogonal nomenclature, feature, sequence, protein, and assay namespaces. Chapter 5 explains CRISPR perturbation and guide-pair architectures. Chapter 6 follows RNA molecules through single-cell sequencing into H5AD. Chapters 7 through 9 define and operationalize the MANTRA identifier contract. The final appendices provide compact reference tables, a glossary, and source citations.

2 Biological Foundations

2.1 DNA as a physical molecule and an information-bearing polymer

Deoxyribonucleic acid, or DNA, is a polymer whose monomers contain one of four bases: adenine, cytosine, guanine, and thymine. The bases are conventionally represented by A, C, G, and T. Two antiparallel DNA strands form a double helix through complementary base pairing: adenine pairs with thymine, and cytosine pairs with guanine. The order of bases along a strand encodes sequence information.

A DNA strand has a chemical direction. Its ends are called 5-prime and 3-prime, written 5' and 3', after carbon positions in the deoxyribose sugar. Polymerases synthesize nucleic acids in the 5' to 3' direction. The direction matters whenever a transcript, guide RNA, promoter, coding sequence, or genomic interval is described. A gene located on the reverse genomic strand is read in the opposite genomic direction from a gene on the forward strand, even though both RNA products are synthesized 5' to 3'.

Human nuclear DNA is organized into chromosomes. Most somatic human cells are diploid: they carry two homologous copies of each autosome, one inherited from each parent, along with a pair of sex chromosomes. The two homologous chromosomes are similar rather than byte-identical. They can contain single-nucleotide variants, insertions, deletions, copy-number differences, and larger structural differences. A reference genome is therefore a coordinate scaffold and representative sequence, rather than the complete genome sequence of every individual cell.

DNA is packaged with proteins into chromatin. Nucleosomes wrap DNA around histone proteins, and higher-order chromatin organization affects whether regulatory machinery can access a region. This physical packaging is central to transcription and explains why chromatin measurements from projects such as ENCODE can provide useful priors for perturbation modeling.

2.2 Genome, assembly, chromosome, contig, and coordinate

The *genome* is the complete genetic material of an organism or cell. A *genome assembly* is a computationally and manually curated representation of that genome built from sequenced fragments. An assembly contains chromosome sequences, unplaced or unlocalized contigs, patch sequences, and metadata. GRCh38 is the thirty-eighth major human reference assembly produced through the Genome Reference Consortium lineage. The suffix in GRCh38.p14 denotes patch release 14.

A genomic interval is usually represented by chromosome, start, end, and strand. Those values are meaningful in an assembly-specific coordinate system. The same biological locus can have different numerical coordinates on GRCh37 and GRCh38. Coordinate-bearing resources therefore carry an assembly identity. Annotation files and chromatin-signal files are combined through their common assembly and coordinate conventions.

The coordinate system also has an indexing convention. Some formats use one-based inclusive intervals; others use zero-based half-open intervals. GTF files use one-based inclusive coordinates. BED files conventionally use zero-based half-open coordinates. The assembly and interval convention together determine how numerical coordinates are interpreted.

2.3 The historical and modern meanings of a gene

The gene began as an abstract unit of inheritance. Classical genetics inferred genes from patterns of trait transmission before the molecular substrate was known. Molecular biology later associated genes with DNA sequences that encode RNA and, for many genes, proteins. Modern genomics has made the definition more relational. A gene is an annotated locus associated with one or more RNA products or other recognized functional outputs, together with regulatory and historical context.

The simple slogan “one gene, one protein” is useful as history but inadequate as a data model. One gene can have multiple promoters, multiple transcription start sites, alternative splice forms, alternative 3' ends, and several protein isoforms. Many genes produce functional RNAs without producing proteins. Some annotated loci are pseudogenes whose evolutionary history connects them to protein-coding genes while their present coding capacity differs. Some readthrough transcripts span neighboring genes. Some regulatory effects arise from the act of transcription rather than from a stable RNA product.

For identifier engineering, a gene is best treated as a locus-level biological entity represented by one or more database records. The gene record can carry a symbol, name, genomic location, locus type, cross-references, and relations to transcript records. The transcript records then represent the alternative RNA structures produced from that locus.

2.4 Locus boundaries, promoters, enhancers, and regulatory regions

A gene locus occupies a genomic region, but its biologically relevant regulatory architecture extends beyond the bases incorporated into mature transcripts. A promoter is a region near a transcription start site where transcription factors and RNA polymerase assemble. A gene can have several promoters, each initiating a different family of transcripts. Enhancers are regulatory elements that can increase transcription from a promoter, often across substantial genomic distances. Silencers, insulators, and other cis-regulatory elements contribute additional control.

The word *cis* means that a regulatory element acts through its position on the same DNA molecule or chromosome context. The word *trans* refers to diffusible factors, such as transcription-factor proteins or regulatory RNAs, that can act on other loci. These concepts matter for perturbation labels. A guide can target a promoter, a distal enhancer, an exon, a splice site, or an entire gene-level intervention. The intended biological object is established by the library design rather than inferred from a gene symbol alone.

2.5 Transcription: synthesis of RNA from a DNA template

Transcription is the synthesis of RNA using DNA as a template. RNA polymerase and associated transcription factors assemble near a promoter, open a local DNA region, and synthesize an RNA strand complementary to the template strand. The process is conventionally divided into initiation, elongation, and termination.

For many human protein-coding genes, the first RNA product is a primary transcript or pre-mRNA. The primary transcript can contain exons, introns, untranslated regions, and a protein-coding region. It is processed before becoming a mature messenger RNA. Processing commonly includes addition of a 5' cap, cleavage and polyadenylation at the 3' end, and splicing.

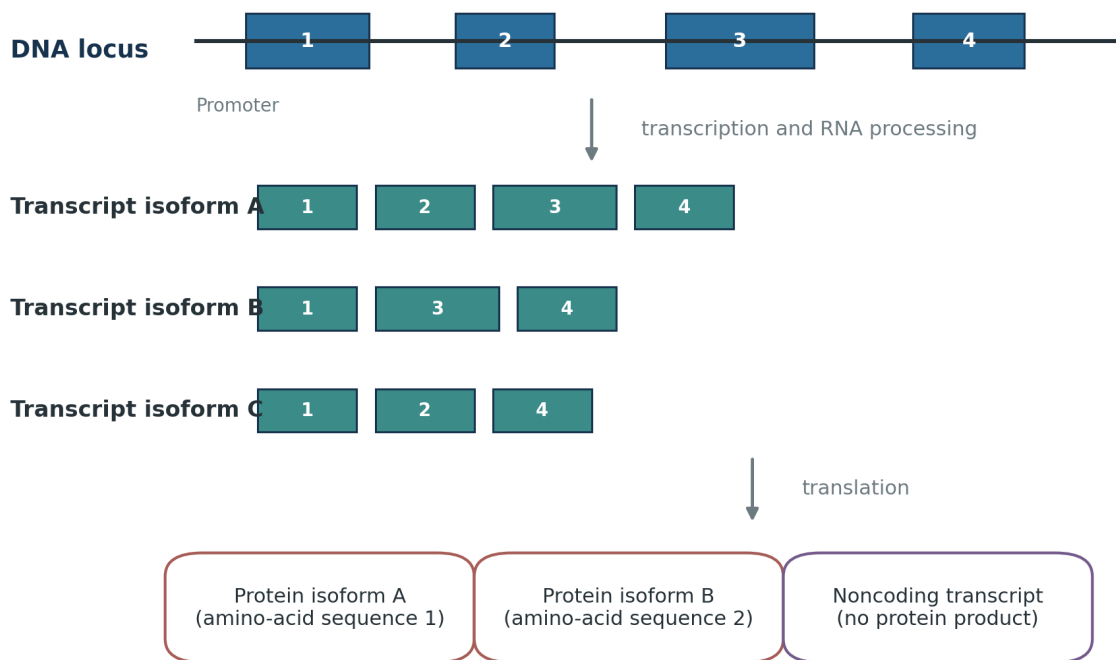
An exon is a segment represented in the mature transcript after splicing. An intron is a transcribed segment removed during splicing. Exons can contain protein-coding sequence, untranslated sequence, or

both. The first exon of a transcript can be entirely untranslated. The last exon can include the end of the coding sequence and a long 3' untranslated region. Consequently, “exon” and “protein-coding segment” are different concepts.

2.6 RNA processing and alternative splicing

Splicing removes introns and ligates selected exons. The spliceosome recognizes sequence and structural signals around splice sites. Alternative splicing allows a locus to produce more than one mature transcript. Common patterns include exon skipping, alternative 5' splice sites, alternative 3' splice sites, mutually exclusive exons, and intron retention. Alternative promoters and alternative polyadenylation sites further expand transcript diversity even when the internal splice pattern is unchanged.

From genomic locus to transcript and protein isoforms



One genomic locus can produce several transcript structures and protein outcomes.

A transcript isoform is one particular mature RNA structure associated with a gene. The structure is defined by an ordered set of exonic intervals on a strand, together with transcript boundaries and, when applicable, a coding sequence. Two transcripts can differ by a single splice junction, a different start site, a different terminal exon, or a different retained intron. An annotation database assigns each modeled transcript its own transcript identifier.

The term *transcript model* emphasizes that an annotation record is a scientific representation inferred from evidence. The model specifies the transcript’s genomic structure and attributes. The physical RNA molecules in cells form a population with variable abundance, processing states, and sometimes additional heterogeneity. The annotation model provides a stable reference object that sequencing reads can be assigned to and researchers can discuss.

2.6.1 Exon skipping

In exon skipping, one transcript includes an internal exon while another connects the preceding exon directly to the following exon. The choice changes the mature RNA sequence. When the skipped exon intersects the coding sequence, the protein can lose a segment or experience a reading-frame change.

2.6.2 Alternative splice sites

A splice donor or acceptor can be selected at more than one nearby position. The resulting transcripts differ by a portion of an exon. Small boundary changes can add or remove amino acids, alter a protein domain, change an untranslated region, or introduce a premature stop codon.

2.6.3 Intron retention

An intron retained in the mature RNA becomes part of the transcript sequence. Retention can regulate transcript localization, stability, or translation. It can also produce a nonproductive transcript that is degraded through quality-control pathways.

2.6.4 Alternative transcript starts and ends

A gene can initiate transcription from several promoters and terminate at several polyadenylation sites. These choices can change the first exon, 5' untranslated region, coding start, last exon, or 3' untranslated region. Alternative ends are part of transcript identity even when most internal exons are shared.

2.7 Isoforms: a precise distinction among transcript, protein, and modification states

The word *isoform* is widely used and can hide several distinct mechanisms. A transcript isoform is an RNA structure. A protein isoform is a distinct protein sequence or proteoform associated with the same gene. Transcript differences can produce protein differences, with a relation determined by coding sequence, reading frame, translation, and downstream processing.

A gene can have several transcript isoforms that encode the same amino-acid sequence because their differences occur only in untranslated regions. A transcript can be annotated as noncoding and therefore produce no protein. Two coding transcripts can use different start codons or reading frames. A single protein sequence can also undergo post-translational modifications such as phosphorylation, glycosylation, acetylation, or proteolytic cleavage. Those modified molecular forms are often called proteoforms and are conceptually downstream of the transcript-derived protein sequence.

Term	Primary object	Source of variation	Typical identifier
Transcript isoform	mature RNA structure	promoter choice, splicing, transcript end, retained intron	Ensembl ENST, RefSeq NM_ or NR_
Protein isoform	amino-acid sequence associated with a gene or entry	alternative coding transcript, start site, or processing	UniProt accession with isoform suffix, Ensembl ENSP, RefSeq NP_
Proteoform	chemically specific protein molecule	post-translational modification, cleavage, sequence variant	specialized proteomics representation; often beyond a gene-ID table

A useful data model therefore stores the gene, transcript, and protein as separate object types connected by explicit relations. It also records the annotation or sequence release that defines the relation. The statement “this is the protein for this gene” is shorthand for a chain such as gene record → coding transcript model → translated sequence → protein entry.

Concept box - Canonical denotes a policy-selected representative.

Annotation and protein databases often choose a representative or canonical transcript because many applications need one default. The selected transcript is an operational reference chosen by a defined policy. Other transcript isoforms remain legitimate biological and annotation objects.

2.8 Translation: from coding RNA to protein

Translation is the synthesis of a polypeptide by the ribosome using a messenger RNA as the template. The ribosome reads nucleotide triplets called codons. Most codons specify amino acids; stop codons terminate translation. The coding sequence begins at a selected start codon and ends at a stop codon. The untranslated regions on either side can influence RNA stability, localization, and translation efficiency while remaining outside the amino-acid sequence.

A protein is a polymer of amino acids that folds into a three-dimensional structure and can participate in catalysis, signaling, transport, structural support, regulation, and many other functions. Protein databases integrate sequence, function, domains, subcellular localization, interactions, disease associations, and evidence. This is why a UniProt accession is a protein identity rather than a gene identity.

The mapping from transcript to protein is partial. Protein-coding transcripts can have translated products. Noncoding transcripts have no ordinary protein product. Some transcript models are retained as nonproductive or subject to nonsense-mediated decay. Pseudogene transcripts can occasionally have translation evidence, but that evidence is annotated separately from the locus’s gene classification.

2.9 Coding and noncoding as annotation properties

A protein-coding gene is a locus with at least one transcript whose evidence and sequence support a protein product. A noncoding RNA gene is primarily annotated for an RNA product rather than a conventional protein product. The word *noncoding* describes coding potential in the annotation, while biological activity can include RNA structure, regulation, localization, and molecular interactions. Noncoding RNAs can contribute to ribosome structure, RNA processing, chromatin regulation, transcriptional control, RNA stability, translation, and many other processes.

The coding status can differ at the transcript level within a gene. One gene record may contain coding transcripts and noncoding transcripts. A transcript can lose coding potential because an alternative splice pattern disrupts the open reading frame or introduces a premature termination codon. Coding status is therefore attached to specific transcript models as well as summarized at the gene level.

2.10 Major classes of noncoding RNA

Noncoding RNAs span several size scales and biological roles.

Class	Approximate role or defining property	Representative identifier context
rRNA	structural and catalytic core of ribosomes	RNA gene and sequence records

Class	Approximate role or defining property	Representative identifier context
tRNA	adaptor between codons and amino acids	tRNA gene annotations
snRNA	spliceosomal and nuclear RNA-processing functions	small nuclear RNA gene records
snoRNA	modification and processing of other RNAs, especially rRNA	small nucleolar RNA gene records
miRNA	short regulatory RNA derived from a precursor hairpin	HGNC gene record, miRNA-specific resources
lncRNA	operational class of noncoding transcripts commonly longer than 200 nt	HGNC, Ensembl, GENCODE, RefSeq
antisense RNA	transcript overlapping another locus on the opposite strand	annotation relation and genomic context
enhancer-associated RNA	transcript associated with active regulatory elements	experimental and annotation context

The categories can overlap. A long transcript can be antisense to a protein-coding gene. A lncRNA locus can overlap an enhancer-like regulatory region. A pseudogene can be transcribed into a noncoding RNA. The identifier record should preserve the database's exact locus and transcript biotypes rather than infer one class from the symbol spelling.

2.11 Long noncoding RNAs in depth

Long noncoding RNA, abbreviated lncRNA, is an operational category. The conventional boundary of roughly 200 nucleotides arose partly from laboratory fractionation practices that separated longer RNAs from well-known small RNAs. It is an operational convention rather than a mechanistic boundary. The category contains heterogeneous transcripts with different genomic positions, structures, evolutionary conservation, expression patterns, and functions [18].

A lncRNA can be intergenic, meaning its locus lies between annotated protein-coding genes; antisense, meaning it overlaps another gene on the opposite strand; intronic, meaning it lies within an intron of another gene; or bidirectional, meaning its transcription initiates near another promoter in the opposite direction. GENCODE and HGNC use controlled biotypes and locus types to encode these relationships [2,6].

Several mechanisms can make a lncRNA locus biologically consequential:

- The RNA molecule can bind proteins, DNA, or other RNAs and act as a scaffold, guide, decoy, or regulatory complex component.
- Transcription through the locus can alter local chromatin or interfere with nearby transcription.
- Splicing or promoter activity can influence neighboring regulatory architecture.
- The RNA can modulate stability, localization, or translation of other transcripts.
- Some loci classified as lncRNA may contain short or noncanonical open reading frames, while the primary annotation remains noncoding.

These possibilities explain why a gene-level perturbation of a lncRNA locus needs experimental context. CRISPR interference near the promoter can suppress the act of transcription and the RNA product together. An RNA-targeting perturbation can affect the RNA molecule more directly. A nuclease-induced indel inside a noncoding exon may have little effect unless it disrupts an important motif. The perturbation identifier alone therefore benefits from method and target-region metadata.

2.11.1 lincRNA, lncRNA, and related labels

The term *lincRNA* means long intergenic noncoding RNA and describes a genomic-context subclass. The broader term *lncRNA* includes intergenic, antisense, intronic, and other long noncoding transcripts. Database biotype vocabularies have changed over time, and a current release may use a more consolidated terminology than an older release. The exact biotype string should be read in the context of the annotation release that issued it.

2.11.2 Expression and evidence

Many lncRNAs are expressed at lower abundance or in narrower cell-state ranges than widely expressed protein-coding genes. Short-read data can provide splice-junction and expression evidence, while long-read sequencing can reveal full-length structures. Low abundance, repetitive sequence, overlapping transcription, and incomplete tissue sampling make lncRNA annotation an active area of refinement. This is one reason release pinning is central to transcript-level identity.

2.12 cDNA: complementary DNA as an experimental representation of RNA

Complementary DNA, abbreviated cDNA, is DNA synthesized from an RNA template by reverse transcriptase. The “complementary” name reflects base pairing: the DNA sequence synthesized by the enzyme is complementary to the RNA template. The process is called reverse transcription because information flows from RNA into DNA, opposite the direction of ordinary cellular transcription [20].

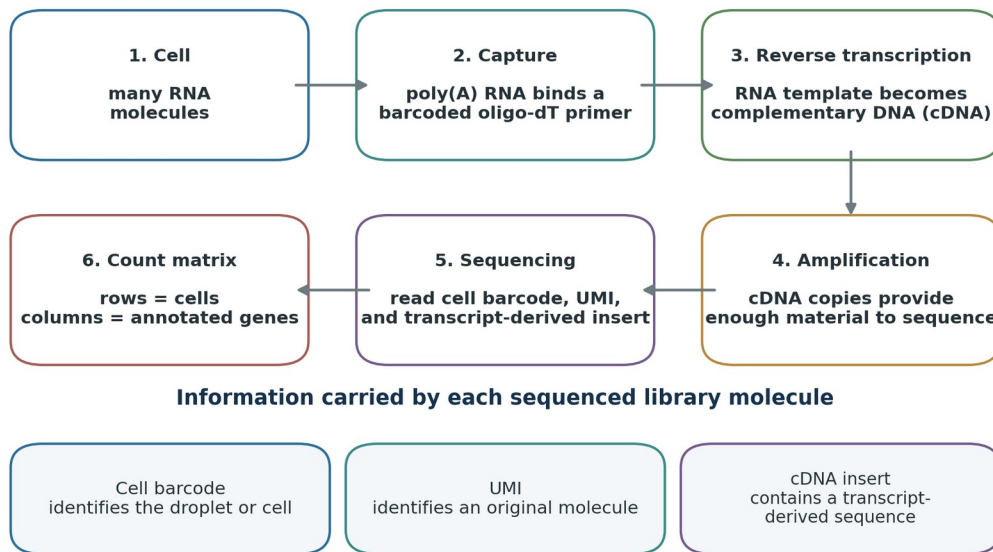
A cDNA molecule represents the RNA that served as its template. When the template is a mature spliced messenger RNA, the cDNA usually joins the exon-derived sequence and lacks the genomic introns that were removed from the RNA. This property historically made cDNA clones powerful evidence for transcript structures. A genomic DNA sequence contains the introns and intergenic context; a mature-RNA-derived cDNA contains the sequence of the processed RNA captured by the experiment.

The term cDNA is used at several scales:

- A full-length cDNA can represent nearly the complete mature transcript.
- A partial cDNA can cover only part of a transcript.
- A cDNA library is a collection of cDNA molecules made from an RNA population.
- In single-cell RNA sequencing, barcoded cDNA molecules are the amplifiable intermediate from which sequencing libraries are built.
- In reference-sequence contexts, “coding DNA” can also appear in variant nomenclature, where it refers to coordinates on a coding transcript reference. That use is related but distinct from a laboratory cDNA molecule.

RNA is chemically less stable than DNA, while ordinary DNA polymerases amplify DNA templates. Reverse transcription converts captured RNA molecules into DNA, which can then be amplified and sequenced. Single-cell workflows add cell barcodes and unique molecular identifiers during or around first-strand cDNA synthesis so that sequenced reads retain information about the cell and original molecule from which they arose [16,17].

Why single-cell RNA sequencing measures cDNA rather than RNA directly



Reads are mapped to an assembly and annotation, then collapsed by cell barcode, gene assignment, and UMI.

RNA is reverse-transcribed into barcoded cDNA before amplification and sequencing.

2.12.1 cDNA, genomic DNA, and transcript models

The three should be distinguished precisely.

Object	Physical or computational status	Contains introns?	Typical use
Genomic DNA	physical chromosome sequence or assembly representation	yes, where the locus has introns	genomic coordinates, guide design, regulatory context
cDNA	physical DNA copy synthesized from RNA	usually reflects the processed RNA and therefore omits spliced introns	sequencing, cloning, transcript evidence
Transcript model	database record describing an inferred RNA structure	represents exon structure and transcript sequence	annotation, read assignment, transcript identifiers

A transcript model can be supported by many cDNA molecules and other evidence. A complete transcript model integrates many cDNA observations with splice-junction, long-read, protein, and curation evidence. Annotation integrates multiple observations, splice-junction evidence, long-read evidence, protein evidence, and curation.

2.13 Pseudogenes in depth

A pseudogene is a gene-like genomic locus with substantial homology to a functional ancestral gene and a present classification that differs from an ordinary protein-coding gene. HGNC describes pseudogenes in

terms of homology and inability to produce the ancestral functional protein product, while GENCODE assigns detailed biotypes reflecting evolutionary origin, transcription, and other evidence [6,19].

Pseudogenes arise through several evolutionary routes.

2.13.1 Duplicated or unprocessed pseudogenes

A genomic segment containing a functional gene can be duplicated. One copy continues to experience selection for the original function, while the other copy can accumulate substitutions, insertions, deletions, frameshifts, or premature stop codons. Because the duplication copied genomic DNA, the pseudogene can retain an exon-intron organization resembling the parent gene. This class is often called duplicated or unprocessed.

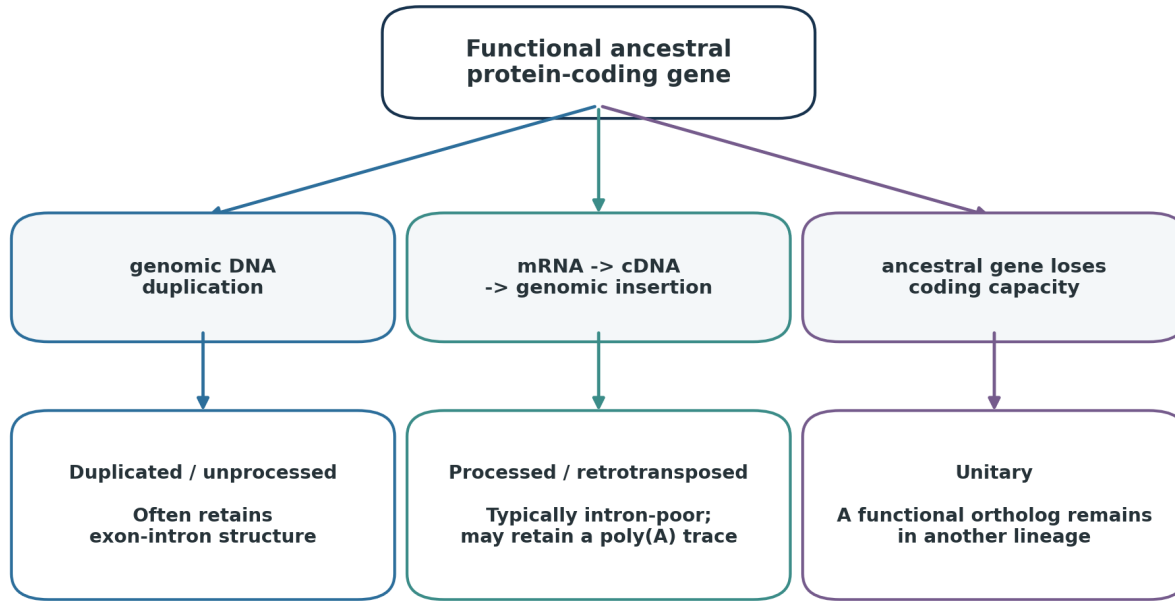
2.13.2 Processed pseudogenes

A mature messenger RNA can be reverse-transcribed into cDNA and inserted into the genome through retrotransposition. The inserted sequence reflects the processed RNA, so it usually lacks the parent gene's introns and can retain signs of a poly(A)-derived tail. The insertion initially lacks the original promoter and regulatory context, although it may acquire transcriptional activity from its genomic neighborhood. This class is called processed or retrotransposed.

2.13.3 Unitary pseudogenes

A unitary pseudogene arises when the original functional gene in one lineage accumulates disabling changes without a retained functional duplicate in that same lineage. A functional ortholog in another species provides evidence that the ancestral locus was protein-coding. The human locus is then classified as unitary when the human lineage carries the disrupted form.

Three major evolutionary routes to pseudogene loci



Origin and present genomic structure define the pseudogene class.
Transcriptional activity and protein-coding capacity remain separately annotated properties.

Pseudogene classes encode evolutionary origin and present structure.

2.13.4 Transcribed and translated pseudogene evidence

Pseudogene status and transcriptional activity are separate properties. Some pseudogene loci are transcribed. Their RNAs can be by-products of local transcription, or they can have regulatory effects. GENCODE includes biotypes such as transcribed processed pseudogene and transcribed unprocessed pseudogene. Some loci have mass-spectrometry or other evidence suggestive of translation, which can be annotated separately [6].

This separation matters for computational priors. A locus classified as a pseudogene can have an RNA-expression profile and chromatin context. A protein-sequence prior requires a defensible current protein product and accession. The typed identity record can represent the locus, transcript evidence, and protein coverage independently.

2.13.5 Why pseudogene symbols often resemble parent-gene symbols

HGNC pseudogene nomenclature frequently includes a P marker or a parent-derived name. A processed pseudogene can receive a parent symbol plus P and a number. An unprocessed family-member pseudogene can carry a family-style symbol ending in P. The spelling communicates lineage and family relationship, while the stable HGNC ID identifies the nomenclature record [1,19].

2.13.6 Pseudogenes in expression data

Pseudogenes can be difficult to quantify because their sequences may be highly similar to parent genes or paralogs. Short reads can map ambiguously. An expression matrix can include a pseudogene column if

the annotation and read-assignment procedure support it. The column's gene symbol, Ensembl ID, and unique matrix key remain separate pieces of metadata.

2.14 Paralogs, orthologs, and gene families

A gene duplication creates paralogs: related genes within a genome or lineage. Speciation creates orthologs: genes in different species descended from a gene in their last common ancestor. Gene families contain homologous genes connected by duplication and divergence. Pseudogenes can also have parent-gene and family relations.

These relationships motivate another reason symbols alone are insufficient for machine joins. Family members can have similar symbols and sequences. Cross-species resources can use species-specific Ensembl prefixes or nomenclature authorities. A human gene identity should therefore carry its species or taxonomy context alongside the namespace.

2.15 A compact biological object graph

The biological relationships developed in this chapter can be summarized as a graph:

$$\text{assembly interval} \supseteq \text{gene locus} \rightarrow \{\text{transcript models}\} \rightarrow \{\text{protein sequences}\}.$$

Regulatory elements connect to promoters and genes through cell-type-dependent relations. Experimental perturbations target positions or mechanisms within this graph. Sequencing assays observe molecules and map the observations back onto annotation features. Identifier systems provide typed names for the nodes, edges, releases, and experimental records.

3 Genome Annotation

3.1 Assembly and annotation solve different problems

A genome assembly provides the reference sequence and coordinate system. A genome annotation provides a structured interpretation of that sequence. The assembly answers, “What bases occupy this reference coordinate?” The annotation answers, “Which biological features are represented at these coordinates, and how are those features organized into genes, transcripts, exons, coding sequences, and other records?”

The distinction is foundational. GRCh38.p14 identifies a human assembly state. GENCODE release 44 identifies an annotation package constructed on that assembly and synchronized with Ensembl release 110. A chromatin-signal file can also be aligned to GRCh38. The common assembly permits coordinate integration, while GENCODE supplies the gene and transcript features used to summarize the signal.

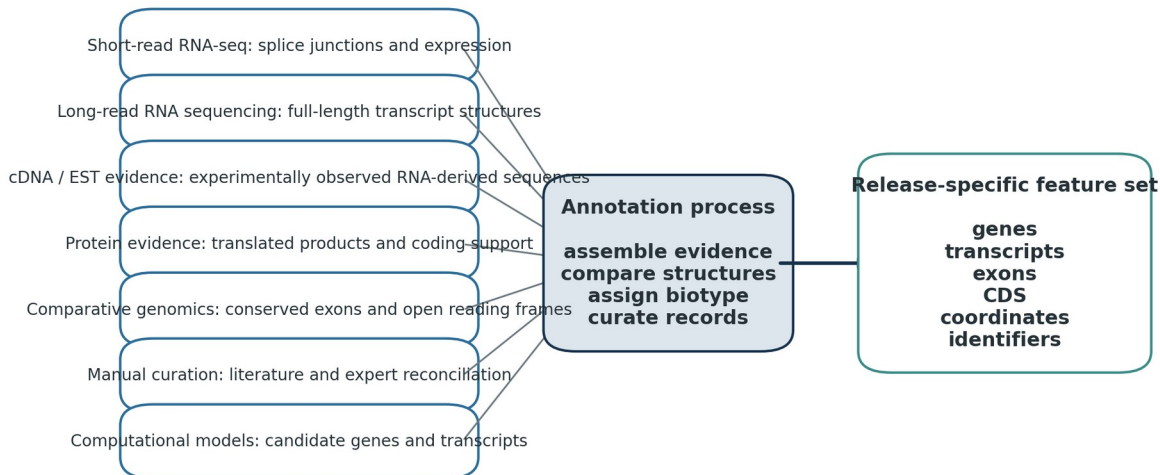
An assembly can remain fixed while annotation releases continue to evolve. New RNA evidence, improved algorithms, manual curation, and cross-database reconciliation can change the gene set without changing the underlying chromosome sequence. Conversely, an assembly change can move coordinates even when the biological interpretation of a gene remains similar.

3.2 What an annotation team actually constructs

An annotation team receives a reference assembly and many forms of evidence. The team or pipeline identifies candidate transcribed regions, splice junctions, transcript starts and ends, open reading frames, protein homology, pseudogene signatures, noncoding RNA structures, and other features. It then produces a coherent, nonredundant set of gene and transcript models.

The resulting annotation is a graph of features rather than a flat list of symbols. A gene feature contains or is associated with transcript features. Each transcript contains an ordered set of exon features. Coding transcripts contain coding-sequence segments and translated protein products. Attributes connect records to HGNC IDs, gene symbols, transcript names, biotypes, support levels, tags, and cross-references.

Genome annotation is evidence integration over a reference assembly



The release records a defensible scientific state, rather than a timeless discovery of a single final answer.

Annotation integrates heterogeneous evidence into a release-specific feature set.

3.3 Sources of annotation evidence

3.3.1 Short-read RNA sequencing

Short-read RNA sequencing produces many fragments derived from RNA molecules. Reads that cross exon-exon junctions support splice connections. Read coverage supports expression and transcribed intervals. Short reads are powerful at scale, but reconstructing complete transcript structures from fragments can be ambiguous when a locus contains many exons and isoforms.

3.3.2 Long-read RNA sequencing

Long-read technologies can sequence much larger portions of individual RNA or cDNA molecules, sometimes approaching full-length transcripts. They can connect distant splice junctions in a single observation and thereby resolve combinations of exons. Long reads also have platform-specific error profiles and sampling limitations, so annotation integrates them with other evidence.

3.3.3 Full-length cDNA and expressed-sequence tags

A full-length cDNA clone provides direct evidence for a processed RNA sequence. Expressed-sequence tags, historically called ESTs, are shorter cDNA-derived sequence observations. These resources were especially important before modern high-throughput RNA sequencing and remain part of the evidentiary history of many transcript records.

3.3.4 Protein evidence

A transcript with an open reading frame can be compared with known proteins, conserved domains, and mass-spectrometry observations. Protein evidence helps determine coding potential and coding boundaries. Because protein databases and genome annotations have different objects and release cycles, the evidence is linked through cross-references rather than represented by one shared identifier.

3.3.5 Comparative genomics

Conservation across species can reveal coding exons, splice sites, regulatory constraints, and ancestral gene structures. Orthologous protein-coding genes can support the interpretation of a human locus. Comparative evidence is also important for identifying unitary pseudogenes, where a functional ortholog exists in another species while the human locus carries disabling changes.

3.3.6 Manual curation and literature

Expert annotators inspect conflicting evidence, published experiments, genomic context, sequence alignments, and database history. Manual curation is particularly valuable in complex loci, gene families, readthrough transcripts, pseudogenes, and clinically important genes. GENCODE combines manual Ensembl-Havana annotation with computational annotation and targeted evidence generation [4].

3.4 Gene models and transcript models

A *gene model* is the annotation's representation of a gene locus and its associated transcript set. A *transcript model* is one release-specific representation of an RNA structure. The model normally includes:

- chromosome or reference sequence;
- strand;
- transcript start and end;
- ordered exon coordinates;
- splice junctions;
- untranslated regions when defined;
- coding-sequence coordinates when applicable;
- gene and transcript biotypes;
- stable accessions and feature versions;
- cross-references and tags; and
- evidence or support attributes.

The word *model* denotes an evidence-based representation of a biological RNA class stored by the database. Physical cells contain individual RNA molecules. The transcript model is the reference object used to group those molecules and interpret sequencing observations.

3.5 Why transcript annotations change

Transcript models change for several reasons. New evidence can reveal an additional exon, a different splice boundary, a previously unobserved transcript start, or a longer 3' end. A sequence error can be corrected. A transcript can be reassigned to another gene after locus boundaries are revised. Two models can be merged, or one model can be split. Coding potential can be re-evaluated. A transcript can be retained while its version increments, or its stable ID can be retired when continuity criteria are no longer met.

Ensembl's stable-ID versioning rules encode entity-specific structural changes. Gene versions increment when the linked transcript set changes. Transcript versions increment for changes in splicing pattern, chromosomal location, or cDNA sequence. Protein versions increment when the peptide sequence changes. Exon versions increment when the exon genomic sequence changes [3].

This entity-specific versioning is why the suffix in ENST00000611639.4 is meaningful. The stable stem ENST00000611639 identifies the transcript record lineage; .4 identifies the fourth feature version under Ensembl's rules. The enclosing Ensembl or GENCODE release identifies the database-wide state in which that version is interpreted.

3.6 Canonical, representative, principal, and MANE transcripts

Many applications require one transcript per gene. Databases therefore provide representative selections. The policy varies by resource and use case. A canonical transcript can be chosen using coding conservation, expression, transcript support, protein evidence, clinical relevance, or other criteria. The choice is a projection from a one-to-many gene-to-transcript relation into a single default record.

MANE, the Matched Annotation from NCBI and EMBL-EBI project, seeks exact agreement between selected RefSeq and Ensembl/GENCODE transcripts for human protein-coding genes. A MANE Select pair gives a matched transcript whose exon structure and sequence agree across the two annotation ecosystems. MANE Plus Clinical provides additional transcripts needed to represent clinically relevant variants. These resources improve interoperability while preserving the separate RefSeq and Ensembl namespaces.

CCDS, the Consensus Coding Sequence project, similarly focuses on agreed protein-coding regions across collaborating resources. CCDS IDs describe consensus coding sequences rather than all transcript features. MANE, CCDS, canonical-transcript tags, and principal-isoform selections are related but distinct constructs.

Concept box - Representative transcript selection is a function of purpose.

A transcript chosen for clinical variant reporting can differ from a transcript chosen for broad expression analysis or protein-domain coverage. A data pipeline should record the selection policy and selected accession rather than assume that the word *canonical* has one global meaning.

3.7 GTF and GFF as annotation exchange formats

GENCODE annotations are commonly distributed in GTF, a tab-delimited format derived from GFF conventions. Each row describes a genomic feature. The standard columns include sequence name, source, feature type, start, end, score, strand, frame, and an attribute field.

GTF column	Meaning
seqname	chromosome or reference sequence
source	producer or annotation source
feature	feature class such as gene, transcript, exon, CDS, start_codon, or stop_codon
start, end	one-based inclusive coordinates
score	optional numeric score
strand	+ or - genomic strand
frame	coding-frame offset for CDS features
attributes	semicolon-delimited identifiers, names, biotypes, tags, and cross-references

A simplified gene row might contain gene_id "ENSG...", gene_name "TP53", and gene_type "protein_coding". A transcript row adds transcript_id "ENST..." and transcript-level attributes. Exon

and CDS rows repeat the parent gene and transcript identifiers so that a parser can reconstruct the hierarchy.

The GTF file is itself a versioned source artifact. A reproducible pipeline records the release, assembly, source URL, byte count, and cryptographic hash. The same GENCODE release can have multiple distributed files, such as comprehensive versus basic annotations or chromosome-specific subsets. The exact file identity determines the records used by the computation.

3.8 Gene and transcript biotypes

A biotype is a controlled annotation category. Gene-level and transcript-level biotypes can differ. A protein-coding gene can contain a retained-intron transcript or another transcript whose coding potential differs from the gene's primary class. Pseudogene biotypes can encode origin and transcriptional evidence. Noncoding loci can be subdivided by genomic context or RNA class.

Biotype vocabularies evolve. A parser should retain the exact source string and, when needed, map it into a project-level ontology through a versioned mapping table. This preserves the original annotation while supporting stable downstream categories.

3.9 Genome assembly patches and alternate loci

A patch release such as GRCh38.p14 adds corrections or alternate representations without redefining every primary-chromosome coordinate. Fix patches provide corrected sequence intended for future incorporation. Novel patches provide alternate representations of structurally variable regions. Assemblies can also include alternate loci, decoy sequences, and unplaced contigs.

The presence of alternate representations means that “the gene's coordinate” can depend on which assembly component is selected. Most expression pipelines operate on the primary assembly or a defined reference bundle. The reference bundle, aligner index, annotation file, and feature-counting configuration together establish the coordinate universe used to build the matrix.

3.10 Coordinate transformations and lift-over

Lift-over maps coordinates between assemblies using alignment chains or related mappings. The transformation can be partial or ambiguous. A region can fail to map, map to several locations, change length, or cross a rearranged interval. Lift-over is therefore a provenance-bearing operation rather than a textual replacement of the assembly label.

For gene-level aggregation, the preferred approach is usually to interpret each source in its native pinned assembly and annotation, then map through stable gene or transcript identities when those mappings are available. Coordinate lift-over remains valuable for signal tracks and intervals, provided the transformation procedure and result quality are recorded.

3.11 Release pinning as scientific state management

A database release is analogous to a versioned state of a large knowledge graph. It contains many records and relations that were mutually consistent under the project's policies at publication time. Reproducibility requires more than an accession string when the accession's attributes and cross-references can evolve.

For the MANTRA annotation layer, the pinned state is:

Field	Resolved value
GENCODE release	44
Ensembl release	110
Genome assembly	GRCh38.p14
Annotation source	GENCODE comprehensive annotation file used by the project
Source-file SHA-256	01f817afed65feee863361b4baf30a95e722ee5c5d508ff77b04106ef7ba20d3

GENCODE's release history records release 44 as a GRCh38.p14 annotation paired with Ensembl release 110 [5]. These values identify three related but orthogonal layers: the annotation package, the genome-database state, and the coordinate assembly.

4 The Identifier Systems

4.1 The central claim: these nomenclatures are orthogonal

HGNC, Ensembl, GENCODE, NCBI Gene, RefSeq, UniProt, ENCODE, genome assemblies, experimental guide labels, and MANTRA coordinate keys are often displayed in one row. Their adjacency can create the impression that they are alternative spellings of one underlying gene ID. A more accurate model is a product space in which each field is a projection onto a different object system.

The major identifier systems answer different questions

HGNC	What is the approved human gene name and stable nomenclature record?	gene nomenclature
Ensembl	Which feature record represents this gene, transcript, exon, or translated protein?	genome feature identity
GENCODE	Which release-specific gene and transcript structures are annotated on this assembly?	annotation state
NCBI Gene	Which NCBI Gene record aggregates this locus and its products?	gene record
RefSeq	Which accessioned reference RNA, protein, or genomic sequence is being used?	reference sequence
UniProt	Which protein knowledgebase entry and protein sequence are being described?	protein knowledge
ENCODE	Which functional-genomics experiment, biosample, or data file produced this signal?	assay provenance
MANTRA	Which experimental row or model response coordinate is being indexed?	project coordinate

Orthogonality means coexistence: one biological case can carry identifiers from every row.

The identifier systems answer different questions and coexist in one typed record.

The word *orthogonal* is used here in the engineering sense. The systems vary along different conceptual dimensions. HGNC primarily standardizes how human genes are named. GENCODE describes which gene and transcript structures exist in a release on an assembly. Ensembl assigns stable identifiers to genomic features and exposes a broader genome database. RefSeq identifies accessioned reference sequences. UniProt identifies protein knowledgebase entries. ENCODE identifies experimental records and files. MANTRA identifies intervention rows and model coordinates.

The systems have cross-references because their objects are related. A cross-reference connects related records while preserving their distinct object identities. An HGNC record can point to an Ensembl gene record. The Ensembl gene can contain many transcripts. A transcript can correspond to a RefSeq RNA

and translate into an Ensembl protein. A UniProt entry can describe the resulting protein and its isoforms. An ENCODE ChIP-seq experiment can measure signal near the gene. Each accession retains the semantics of its issuing namespace.

4.2 A question-to-resource matrix

Question	Primary resource or field	Object returned
What is the approved human gene symbol?	HGNC	nomenclature record and approved symbol
Which stable genome feature is this?	Ensembl ENSG, ENST, ENSP, ENSE	gene, transcript, translated protein, or exon feature
Which gene and transcript structures were annotated in this release?	GENCODE release and GTF	release-specific annotation graph
Which NCBI record aggregates this gene and its products?	NCBI GeneID	NCBI gene record
Which reference RNA or protein sequence is being used?	RefSeq accession.version	sequence record
Which curated protein knowledge entry is being used?	UniProt accession and isoform	protein knowledgebase entry or isoform
Which functional-genomics experiment or file generated this measurement?	ENCODE accession	assay, biosample, file, library, or pipeline record
Which coordinate system defines chromosome positions?	assembly accession/name	reference sequence and coordinate frame
Which experimental intervention row is this?	MANTRA perturbation identity	preserved split-contract row
Which predicted expression coordinate is this?	resolved_gene_key	unique model response coordinate

4.3 HGNC: human gene nomenclature

The HUGO Gene Nomenclature Committee assigns approved human gene symbols and names and maintains stable HGNC IDs. The approved symbol is the concise human-readable label. The HGNC ID is the stable record identifier and remains attached to the record when the approved symbol changes [1,2].

A symbol report includes core nomenclature and contextual fields:

HGNC field	Meaning
HGNC ID	stable identifier for the nomenclature record, such as HGNC:33721
approved symbol	current standardized short label
approved name	current standardized full name
previous symbols	former HGNC-approved symbols for the record
alias symbols	other symbols used in literature or databases

HGNC field	Meaning
locus type	gene class such as protein-coding, long noncoding RNA, or pseudogene
chromosomal location	cytogenetic or genomic location summary
gene group	curated family or functional grouping
cross-references	Ensembl, NCBI Gene, RefSeq, UniProt, CCDS, and other links

HGNC symbols support scientific communication. Their mnemonic value makes them attractive as data labels. The stable HGNC ID supports persistence through nomenclature updates. A reproducible resolver uses a pinned HGNC snapshot and records how an input matched the record: approved symbol, previous symbol, alias, or another defined match class.

4.3.1 Approved symbols, previous symbols, and aliases

An approved symbol is the current HGNC nomenclature. A previous symbol was once approved and has since been replaced for the same or a successor record. An alias is another label associated with the gene through literature or database usage. The fields encode different historical relationships.

A single alias string can occasionally be associated with more than one current record. This is why a resolver returns an explicit candidate set or ambiguity state rather than relying solely on string equality. The HGNC multi-symbol checker and downloadable snapshots provide the information needed for deterministic resolution [2].

4.3.2 HGNC locus types

HGNC covers protein-coding genes, noncoding RNA genes, pseudogenes, immunoglobulin and T-cell-receptor genes, readthroughs, regions, and other genomic features. Locus type is a property of the nomenclature record. It can be used as one input to a protein-coverage rule, while transcript-level coding information remains release-specific annotation data.

4.3.3 Example: ALG1L and ALG1L1P

ALG1L appears in the experimental atlas as a perturbation label. In the pinned HGNC snapshot it resolves as a previous symbol for the approved pseudogene symbol ALG1L1P, record HGNC:33721. The experimental label and current nomenclature are both retained because they answer different questions: which row was used by the experiment, and which current HGNC record provides the nomenclature interpretation.

4.4 Ensembl: stable identifiers for genome features

Ensembl is a genome database and comparative-genomics platform. It assigns stable identifiers to genes, transcripts, translated proteins, exons, and other features. Human feature prefixes encode the object type.

Prefix	Object type	Example
ENSG	gene feature	ENSG00000189366
ENST	transcript feature	ENST00000611639
ENSP	translated protein feature	ENSP00000482705

Prefix	Object type	Example
ENSE	exon feature	ENSE00003979176

The stable stem and version suffix are separate components. In ENSG00000189366.13, the stem identifies the Ensembl gene-record lineage and 13 is the gene-feature version. In ENST00000611639.4, the stem identifies the transcript-record lineage and 4 is the transcript-feature version.

Ensembl stable IDs are designed to be tracked across reannotation. The mapping process compares features between releases and retains stable IDs when continuity criteria are satisfied. The version increments when the entity changes under feature-specific rules [3].

4.4.1 Ensembl gene, transcript, and protein relations

Within a release, one Ensembl gene usually contains one or more Ensembl transcripts. A coding transcript can have an Ensembl protein translation. The relations are release-specific. A transcript can change gene membership, be retired, or receive a new version after structural revision. A gene-to-protein join therefore traverses the release graph rather than substituting prefixes in an accession string.

4.4.2 Species context

Ensembl prefixes can include species codes. Human stable IDs have the familiar ENSG, ENST, and ENSP forms. Mouse identifiers include ENSMUSG, ENSMUST, and ENSMUSP. A robust parser treats prefix structure as a type hint and retains the species or taxonomy context as an explicit field.

4.5 GENCODE: a release-specific human and mouse gene annotation

GENCODE produces high-quality reference annotations for human and mouse genes and transcripts. The annotation is supported by experimental data and combines manual curation, computational analysis, and targeted evidence generation [4]. It is distributed as release files and displayed through Ensembl and the UCSC Genome Browser.

GENCODE is an annotation product rather than a second family of independent gene-accession prefixes. Human GENCODE annotations use Ensembl stable IDs for genes and transcripts. The GENCODE release identifies the coherent annotation package in which those feature records, structures, biotypes, tags, and coordinates are interpreted.

The distinction can be summarized as follows:

Layer	Example	Semantics
Ensembl gene accession	ENSG00000189366	stable gene-feature record lineage
Ensembl feature version	.13	revision of that gene feature
GENCODE release	44	annotation-wide published state
Ensembl release	110	genome-database release synchronized with GENCODE 44
assembly	GRCh38.p14	genomic coordinate system
source file hash	SHA-256 of the GTF	exact bytes used by the computation

4.5.1 Comprehensive and basic annotations

GENCODE distributes multiple annotation subsets. The comprehensive gene set contains the full annotation. The basic set provides a representative subset of transcripts selected by policy. Other files can separate primary-assembly annotation, polyadenylation features, long noncoding RNAs, or metadata. The source filename and hash are therefore part of the identity of an annotation input.

4.5.2 GENCODE tags and transcript support

Transcript records can carry tags describing representative status, support, exceptions, readthroughs, or cross-resource agreement. These tags are metadata within the release. A downstream system can select transcripts by tag while retaining the complete source record and selection rule.

4.6 NCBI Gene: stable gene records in the NCBI ecosystem

NCBI Gene assigns a stable numeric GeneID to each gene record. The historical term *Entrez Gene ID* refers to the same identifier system. NCBI Gene aggregates symbols, names, gene type, genomic locations, RefSeq products, orthology, phenotype and literature links, and cross-references. Human symbols are imported from HGNC when available [7].

A GeneID identifies the NCBI gene record. RefSeq accessions identify the associated transcript and protein sequences. For the current ALG1L1P record, the NCBI GeneID is 200810. RefSeq transcript and protein accessions linked from that record have their own sequence identities.

NCBI tracks record history, including replaced or discontinued records. Reproducible bulk mappings use a pinned NCBI dataset or source export. The GeneID remains a useful cross-resource key because it is stable and widely represented, while the current record attributes can change with database updates.

4.7 RefSeq: accessioned reference sequences

The NCBI Reference Sequence project, RefSeq, provides curated and computationally predicted reference sequences for genomes, transcripts, and proteins. A RefSeq accession identifies a sequence record. The prefix encodes the broad sequence class.

Prefix	Sequence object
NM_	curated or well-supported protein-coding messenger RNA
NR_	curated or well-supported noncoding RNA
NP_	curated or well-supported protein
XM_	predicted protein-coding messenger RNA
XR_	predicted noncoding RNA
XP_	predicted protein
NG_	genomic region or RefSeqGene record
NC_	chromosome or complete genomic molecule in RefSeq

The suffix in an accession such as NM_000546.6 is a sequence version. It increments when the sequence changes. Metadata-only updates can leave the sequence version unchanged. NCBI recommends citing the accession with its version so that the exact sequence is specified [8,10].

RefSeq and Ensembl/GENCODE can describe corresponding transcripts in separate namespaces. A MANE transcript pair represents an especially strong exact match. Outside exact matched sets, mappings can be one-to-one, one-to-many, partial, or absent because the resources use different curation and annotation procedures.

4.8 UniProt: protein knowledge and sequence identity

UniProtKB is a protein knowledgebase. Each entry includes a protein sequence and can include names, function, domains, catalytic activity, subcellular location, post-translational modifications, interactions, disease associations, evidence attribution, and cross-references. UniProtKB/Swiss-Prot contains reviewed, manually annotated entries; UniProtKB/TrEMBL contains unreviewed, computationally annotated entries [7,8].

A primary accession such as P49356 identifies a UniProt protein entry. Isoforms can be represented with suffixes such as P04150-2. The isoform suffix is part of UniProt's protein-level representation and has different semantics from an Ensembl transcript-version suffix.

One gene can be represented by several protein isoforms. A UniProt reviewed entry may consolidate isoforms under one primary accession, while unreviewed entries can have a different organization. A pseudogene may have no current protein accession. The protein identity record is therefore derived through a current gene and transcript interpretation rather than inferred from symbol spelling.

For sequence-derived priors, reproducibility includes the UniProt accession, isoform when applicable, sequence bytes, database release or retrieval date, source location, and SHA-256. The exact FASTA bytes determine the embedding or similarity computation.

4.9 ENCODE: identifiers for functional-genomics data

The Encyclopedia of DNA Elements, ENCODE, is an experimental consortium and data portal. It stores assays, biosamples, donors, libraries, antibodies, genetic modifications, files, and processing pipelines. Its accessions identify those experimental and computational records [9,11].

Prefix	ENCODE object	Example meaning
ENCSR	experiment or assay record	a ChIP-seq, ATAC-seq, RNA-seq, or related experiment
ENCFF	data file	one downloadable raw or processed file
ENCBS	biosample	a specific prepared biological sample
ENCLB	sequencing library	nucleic-acid library associated with an assay
ENCDO	donor	human donor or model-organism strain record
ENCAB	antibody lot	characterized antibody reagent
ENCPL	pipeline	data-processing pipeline record

Prefix	ENCODE object	Example meaning
--------	---------------	-----------------

ENCODE provides measured genomic signal and assay provenance. A peak file or signal track is interpreted against an assembly. Gene-level summarization uses annotation coordinates from GENCODE or another named annotation. The ENCODE file accession, GENCODE release, and assembly occupy separate fields because the assay, feature set, and coordinate frame are different objects.

4.10 Genome Reference Consortium and assembly identifiers

The Genome Reference Consortium lineage provides human assembly names such as GRCh37 and GRCh38. NCBI and INSDC also assign assembly accessions, and chromosome sequences have RefSeq accessions such as NC_000001.11. The assembly name, assembly accession, and component-sequence accessions form a related hierarchy.

For most MANTRA coordinate-derived resources, GRCh38.p14 is the operative assembly label. This identity is paired with the exact GENCODE release and source file. When an external source uses another assembly, the integration record captures the source assembly and any explicit transformation.

4.11 MANE and CCDS as convergence layers

MANE and CCDS illustrate how orthogonal resources can converge without losing namespace identity. A MANE pair links a RefSeq transcript and an Ensembl/GENCODE transcript with matched sequence and structure. The two accessions remain distinct because they are maintained by different systems. The MANE relation asserts equivalence under a specific release and policy.

CCDS identifies consensus protein-coding regions accepted by participating annotation groups. A CCDS ID names the consensus coding-sequence record. It coexists with the parent gene, full-transcript, RefSeq RNA, Ensembl transcript, and UniProt protein identifiers.

4.12 A formal crosswalk model

Let G_H be HGNC gene records, $G_E^{(r)}$ Ensembl gene features in release r , $T_E^{(r)}$ Ensembl transcript features, $S_R^{(s)}$ RefSeq sequence records in snapshot s , and $P_U^{(u)}$ UniProt entries in release u . Cross-resource integration uses relations such as

$$M_{HE}^{(r,h)} \subseteq G_H^{(h)} \times G_E^{(r)},$$

$$M_{ET}^{(r)} \subseteq G_E^{(r)} \times T_E^{(r)},$$

$$M_{TR}^{(r,s)} \subseteq T_E^{(r)} \times S_R^{(s)},$$

and

$$M_{TP}^{(r,u)} \subseteq T_E^{(r)} \times P_U^{(u)}.$$

The superscripts make the release dependence explicit. A pipeline can materialize a convenient table from these relations, but the table's provenance includes the source snapshots and matching procedure.

4.13 Identifier syntax reference

Identifier	Namespace	Object type	Version location
HGNC:33721	HGNC	human gene nomenclature record	snapshot-level context; no feature suffix
ENSG00000189366.13	Ensembl	gene feature	.13 feature version
ENST00000611639.4	Ensembl	transcript feature	.4 feature version
ENSP00000482705.1	Ensembl	translated protein feature	.1 feature version
200810	NCBI Gene	gene record	snapshot-level context
NM_000546.6	RefSeq	RNA sequence record	.6 sequence version
NP_000537.3	RefSeq	protein sequence record	.3 sequence version
P04637	UniProt	protein entry	entry release context
P04150-2	UniProt	protein isoform	isoform suffix -2
ENCSR868FGK	ENCODE	experiment record	portal record state
ENCFF948AFM	ENCODE	file record	file accession and exact downloaded bytes
GRCh38.p14	GRC assembly naming	genome assembly and patch state	p14 patch level
ALG1L	MANTRA experimental contract	perturbation row label	source split contract and atlas provenance
TBCE-1	MANTRA response-axis contract	unique model coordinate	source gene-axis contract

4.14 Orthogonality in one concrete row

A fully resolved row for an experimental perturbation can contain all of the following without redundancy:

```

perturbation_identity: ALG1L
match_kind: previous_symbol
hgnc_id: HGNC:33721
approved_symbol: ALG1L1P
ensembl_gene_id: ENSG00000189366
raw_transcript_label: ENST00000340333.3
current_gencode_transcript: ENST00000611639.4
ncbi_gene_id: "200810"
locus_type: pseudogene
uniprot_accessions: []
gencode_release: "44"
ensembl_release: "110"
genome_assembly: GRCh38.p14

```

The fields are projections of one case onto experimental, nomenclature, locus, transcript, protein, annotation, and coordinate systems. The empty protein list is informative because it states the resolved coverage status of the protein namespace.

5 Perturbation Experiments and Guide Architecture

5.1 Perturbation as an experimental object

A perturbation is an intervention applied to a biological system in order to alter a molecular process and observe the resulting phenotype. Perturbations can be genetic, chemical, environmental, or physical. In the present data stack, the central perturbations are CRISPR-based genetic interventions whose target labels are associated with genes and transcripts.

The experimental identity is more specific than the biological target name. “Perturb TP53” can refer to a nuclease knockout, promoter repression by CRISPR interference, promoter activation by CRISPR activation, transcript degradation, base editing, or a pair of guides designed to delete a region. Each method acts through a different mechanism and can produce a different distribution of molecular outcomes. The perturbation record therefore includes the target label, guide or guide pair, construct architecture, effector system, and library context.

5.2 CRISPR-Cas targeting from first principles

CRISPR systems use a guide RNA to direct a Cas protein or Cas-derived effector to a DNA sequence through base pairing. A commonly used single-guide RNA, or sgRNA, contains a variable targeting region and a constant scaffold region. The targeting region is designed to be complementary to a genomic sequence adjacent to a protospacer-adjacent motif, abbreviated PAM, recognized by the Cas protein.

The guide RNA and Cas protein form a complex. The guide targeting sequence base-pairs with genomic DNA at compatible sites. The PAM and guide-target complementarity determine candidate binding. The downstream biological effect depends on the effector:

- Active Cas9 can introduce a double-strand break.
- Catalytically inactive Cas9, called dCas9, can bind DNA without cutting it.
- dCas9 fused to transcriptional repressor domains can implement CRISPR interference, or CRISPRi.
- dCas9 fused to activator domains can implement CRISPR activation, or CRISPRa.
- Base editors and prime editors use guide-directed binding to implement other sequence changes.

A guide identity is consequently an experimental sequence and construct record. Its gene assignment is derived from guide sequence, reference assembly, target region, and library design. Gene targeting is an annotation relation derived from the guide sequence, reference assembly, target region, and library design.

5.3 CRISPR nuclease knockout

In a conventional Cas9 knockout experiment, the guide directs nuclease-active Cas9 to a genomic site. Repair of the double-strand break can introduce insertions or deletions. When the target lies in a coding exon, some repair outcomes disrupt the reading frame or important protein sequence. The population can contain several alleles and editing outcomes.

The intended perturbation label is often a gene, while the physical target is a genomic protospacer. The outcome depends on guide efficiency, repair distribution, ploidy, copy number, and whether all functional alleles are disrupted. A gene-level label summarizes the intervention intent; the guide and target coordinates describe the actual design.

5.4 CRISPR interference

CRISPRi uses catalytically inactive Cas9 fused to a transcriptional repressor, such as a KRAB-containing effector. The sgRNA directs the complex to a promoter or transcription-start-site region. The bound effector recruits repressive chromatin machinery and reduces transcription. CRISPRi therefore produces knockdown rather than a DNA-cutting genotype.

The mechanism has several consequences for identity. The target region is often promoter-specific. A gene with several promoters can have promoter-dependent effects. A transcript label in a library can encode the promoter or transcript context used during guide design. A historical transcript assignment can remain important experimental metadata even when a newer annotation release reorganizes the locus.

The Replogle genome-scale Perturb-seq experiments used CRISPRi to target expressed genes across millions of human cells [13]. The associated dual-sgRNA library work showed that two guides targeting the same gene can increase knockdown while allowing a compact library architecture [14,15].

5.5 CRISPR activation

CRISPRa uses dCas9 fused or recruited to transcriptional activators. Guides are designed near promoters or other regulatory regions to increase expression. The perturbation label can again be gene-level while the physical target is a promoter-region sequence. A guide pair can increase activation efficacy or target alternative promoters.

5.6 The semantic layers of a guide record

A complete guide record commonly contains:

Field	Meaning
guide ID	library-local identifier for the sgRNA
protospacer sequence	variable sequence that directs targeting
scaffold or constant-region design	sgRNA architecture and capture compatibility
target assembly	genomic coordinate system used for design
target coordinate and strand	intended genomic binding site
PAM	adjacent motif required by the Cas system
target gene or transcript label	library-design interpretation
predicted activity	score estimating efficacy
off-target metrics	scores or candidate alternative sites
vector position	A or B position in a dual-guide cassette
construct ID	identity of the complete single- or paired-guide library element

The guide ID, target gene, transcript label, and construct ID are therefore distinct. Two guides can share a gene target. The same guide can appear in different paired constructs. Position A and position B can have different constant regions or promoters even when their targeting sequences are functionally symmetric.

5.7 Single-guide library elements

A single-guide library element encodes one sgRNA. The element can be delivered to cells with a lentiviral vector or another system. In a pooled screen, many cells receive different library elements. The abundance or phenotype of cells associated with each guide is then measured.

A gene can be represented by several independent single-guide elements. Multiple guides increase the chance that at least one guide has strong activity and allow agreement across guides to support the gene-level effect. The cost is a larger library and a larger number of cells required to maintain coverage.

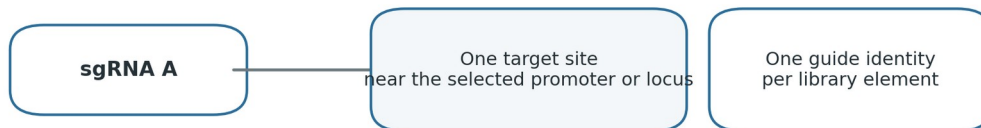
For single-cell screens, the guide identity can be captured directly, inferred through a linked guide barcode, or assigned through another indexing strategy. The cell-level record associates one or more guide identities with the cell barcode.

5.8 Dual-guide elements targeting one gene

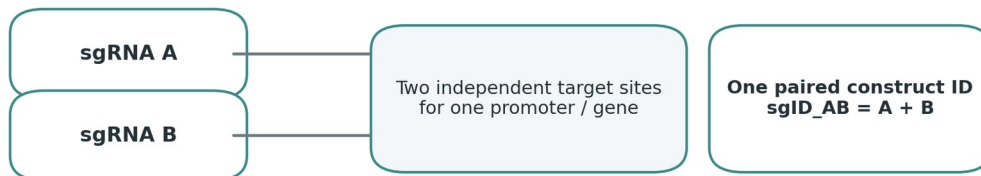
A dual-guide element encodes two sgRNAs, commonly called guide A and guide B, in one construct. When both guides target the same gene, they bind two distinct target sites associated with the gene's promoter or regulatory region. The two guide activities can combine to produce stronger or more consistent transcriptional repression than one guide alone.

Single-guide and paired-guide CRISPR library architectures

A. Single-guide element



B. Dual-guide element targeting the same gene



C. Other paired-guide uses



The phrase “dual guide” describes construct architecture; the biological target relation must be read from the library design.

Single-guide and paired-guide constructs support several experimental designs.

The central distinction is between *two guides per construct* and *two genes per perturbation*. A same-gene dual-guide element contains two sgRNAs whose intended target gene is the same. The biological perturbation remains one gene-level intervention. The exact construct identity retains both guide components.

The Replogle dual-sgRNA CRISPRi design selected pairs of high-performing guides for each gene and expressed them from a tandem cassette. The design enabled one compact library element to carry two targeting sequences. In direct-capture Perturb-seq, both guide positions can be sequenced and assigned at the single-cell level [14,15].

5.8.1 Why two guides can improve knockdown

Guide efficacy varies because of target accessibility, sequence context, distance from the transcription start site, chromatin state, guide expression, and effector occupancy. Two independently selected high-activity guides provide two binding opportunities. Their combined repression can increase the probability and magnitude of target knockdown across cells.

The pair also changes how activity is summarized. A dual-guide construct is a single experimental element with a combined phenotype. The individual guide scores remain useful metadata, while the measured response belongs to the pair as delivered.

5.8.2 Guide A and guide B as positional identities

The A/B labels usually denote positions in the vector cassette. Position can matter for expression promoter, constant region, guide capture, amplification, or recombination analysis. A pair should therefore be represented as an ordered structure when the vector architecture is ordered:

$$q = (q_A, q_B, v, c),$$

where q_A and q_B are guide identities, v is the vector or cassette design, and c is the construct identity. Even when the biological targeting effect is approximately symmetric, the experimental record preserves the positions.

5.9 Paired guides targeting different genes

A dual-guide construct can also target two different genes. This design supports combinatorial perturbation and genetic-interaction mapping. The pair `geneA + geneB` is then a distinct perturbation from either single-gene intervention. Controls often pair a targeting guide with a non-targeting guide so that single-gene effects can be estimated under the same vector architecture.

In a genetic-interaction design, several guide sequences for gene A can be combined with several guide sequences for gene B. The construct-level space is the Cartesian product of selected guide sets. Gene-pair effects are aggregated across constructs while retaining guide-level measurements and positional controls.

This use of paired guides should be distinguished from the same-gene dual-guide architecture in the K562 atlas. The field name `sgID_AB` communicates that the construct contains A and B guide identities; the target metadata determines whether the pair represents two guides for one gene, two genes, a flanking deletion, or another design.

5.10 Paired guides for genomic deletions

Nuclease-active Cas9 can use two guides flanking a genomic region. Cutting at both sites can delete the intervening DNA. This architecture is common for noncoding regulatory elements, exons, or larger segments. Here, the guide pair defines two boundary coordinates, and the intended perturbation object is the interval between them.

The same phrase “dual guide” therefore covers at least three experimental semantics:

Pair architecture	Guide-target relation	Perturbation object
same-gene dual CRISPRi	both guides target one promoter/gene	one gene-level repression
combinatorial pair	guides target two genes or loci	joint intervention and genetic interaction
flanking deletion pair	guides target opposite boundaries of one interval	genomic interval deletion

The construct record resolves the ambiguity by carrying guide targets and library-design type.

5.11 The `sgID_AB` field in the K562 atlas

The raw atlas field `obs/sgID_AB` identifies the paired guide construct assigned to each cell. A displayed ALG1L example has the form:

```
ALG1L_-125655482.23-ENST00000340333.3 |
ALG1L_-125655452.23-ENST00000340333.3
```

The two components correspond to guides A and B. Both components carry the perturbation symbol and transcript-design label, while their coordinate-like portions distinguish the target sites. The complete pair is the exact intervention identity. Cells sharing the gene-level perturbation can therefore be grouped at the perturbation-symbol level while retaining construct-level provenance.

The paired ID is more specific than `obs/gene` and `obs/perturbation_id`. The gene symbol answers which benchmark perturbation row the cell belongs to. The pair answers which physical dual-guide element was present. The transcript label records the design-era transcript or promoter context.

5.12 Direct guide capture

Direct-capture Perturb-seq modifies the guide RNA architecture so that guide-derived sequences can be captured and sequenced alongside the single-cell transcriptome. The guide assignment becomes an observed molecular measurement rather than an inference from a separate linked barcode. Direct capture supports cells containing multiple guides and makes combinatorial or dual-guide designs easier to resolve [15].

At a high level, the output contains two linked measurement streams:

1. gene-expression cDNA reads carrying cell barcodes and UMIs; and
2. guide-derived reads carrying the same cell-barcode context.

The cell barcode joins the streams. The guide assignment establishes the experimental intervention, and the gene-expression matrix establishes the cellular response.

5.13 Linked barcodes and direct capture

Earlier Perturb-seq implementations often associated each guide with a separate guide barcode transcribed in a polyadenylated transcript. The barcode was captured by standard RNA-seq chemistry and mapped back to the guide. Direct capture sequences the expressed guide more directly. Both strategies create an intervention-to-cell relation, but their molecular objects and error modes differ.

A data contract should state which field contains the guide identity, whether the identity was directly captured or inferred, how multiple guides were called, and how uncertain assignments were represented.

5.14 Lentiviral delivery and recombination context

Dual-guide constructs are frequently delivered with lentiviral vectors. Lentiviral particles package two RNA genomes, and reverse transcription during integration can switch templates. In pooled dual-guide libraries, template switching can create recombined A/B pairs. Modern designs and sequencing strategies quantify the observed pair and can exclude or separately analyze recombined constructs. Replogle and colleagues reported direct sequencing of both positions and explicit recombination handling in dual-sgRNA screens [14].

For identifier design, the important principle is that the observed construct identity can differ from the planned library row. The raw guide-pair call should preserve the observed A and B identities. A planned-construct ID and a recombination or validity flag can then describe the relationship to the design table.

5.15 Multiplicity of infection, singlets, and multiple guide assignments

Multiplicity of infection, abbreviated MOI, describes the average number of viral integration events per cell. Low-MOI designs aim to place one library element in most infected cells, which simplifies causal attribution. Some screens intentionally use higher multiplicity or combinatorial constructs. Droplet-based single-cell sequencing can also capture more than one cell in a droplet, creating a multiplet.

A cell can therefore have several guide-derived sequences for several reasons: a designed dual-guide cassette, multiple viral integrations, a recombined construct, ambient guide molecules, or a cell multiplet. Guide-calling software distinguishes these possibilities using sequence counts, expected architecture, quality thresholds, and cell-level metadata. The final cell record should preserve the called construct, component guides, assignment confidence, and multiplicity flags when available.

5.16 Target identity versus response identity

The perturbation target and the measured response genes live on different axes. A guide pair targeting FNTB produces a transcriptional response measured over thousands of genes. The target identity indexes the intervention row. The response-gene identifiers index the expression vector. This asymmetry is the biological basis for MANTRA's separate perturbation and response axes.

5.17 Controls

Non-targeting controls use guide sequences designed to lack a relevant genomic target under the library's criteria. They estimate the baseline effects of vector delivery, guide expression, effector activity, selection, and assay handling. Positive controls target genes with expected phenotypes. Targeting-plus-control pairs can calibrate one side of a paired architecture.

The label non-targeting is an experimental category. It can occupy fields that otherwise carry symbols or gene IDs, but it represents a control intervention rather than a biological gene record. A typed schema assigns it an experimental-control object type.

5.18 Perturbation identity hierarchy

The experimental identities can be organized from broad to specific:

target concept → perturbation label → library element → guide pair → guide sequences and target sites .

For the atlas, the benchmark row is the perturbation label. The exact construct is sgID_AB. The raw transcript field captures design context. The Ensembl gene ID connects the target to a locus record. These layers support different analyses and should remain individually addressable.

6 Single-Cell Perturbation Data

6.1 What Perturb-seq measures

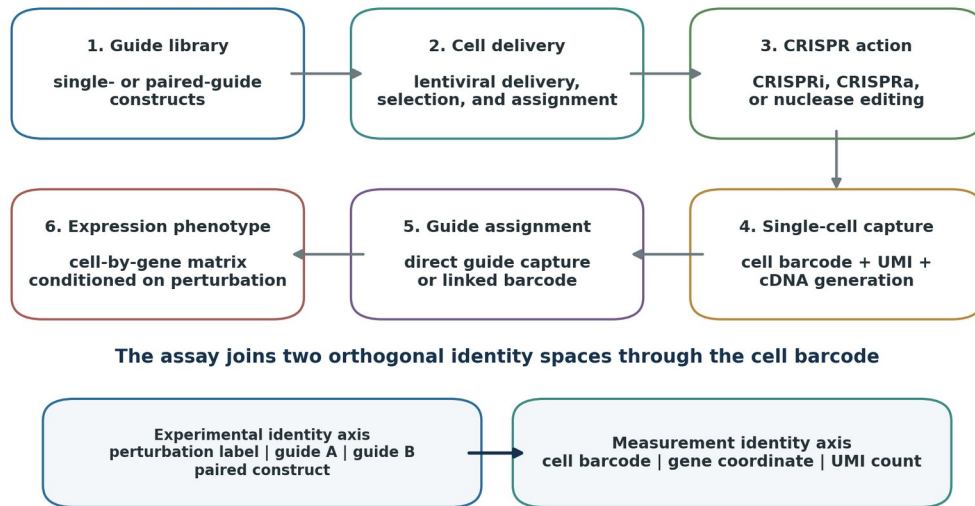
Perturb-seq combines pooled CRISPR perturbations with single-cell RNA sequencing. Each cell has an intervention identity and a transcriptional phenotype. The central output can be viewed as a set of tuples

$$(c_i, q_i, x_i),$$

where c_i is a cell identity, q_i is the called perturbation or guide construct, and x_i is a vector of expression measurements across annotated genes.

The power of the assay comes from the high-dimensional phenotype. Instead of measuring one survival score or reporter, the experiment observes the state of thousands of genes in each cell. Aggregating cells by perturbation yields pseudobulk response profiles, while retaining cells supports heterogeneity, state, trajectory, and distributional analyses.

Perturb-seq links intervention identity to a single-cell transcriptional phenotype



Perturb-seq connects guide identity to a single-cell expression phenotype.

6.2 From cellular RNA to sequencing reads

A droplet-based single-cell RNA-seq workflow begins with a suspension of cells. Microfluidic partitioning places individual cells together with barcoded gel beads and reaction reagents in droplets or emulsions. Messenger RNAs with poly(A) tails hybridize to oligo-dT primers. Reverse transcriptase synthesizes cDNA and incorporates barcode sequences.

A typical molecule carries several information components:

- a sample index identifying the sequencing library or sample pool;
- a cell barcode identifying the droplet or cell partitioning;
- a unique molecular identifier, or UMI, labeling the original captured molecule;

- primer and adapter sequences required for amplification and sequencing; and
- a cDNA insert derived from the captured RNA.

The exact architecture varies by chemistry. In 10x Genomics 3' gene-expression chemistry, cDNA molecules generated in one gel-bead emulsion share a cell barcode, and UMIs distinguish original captured transcript molecules within that cell [16,17].

6.3 Cell barcodes

A cell barcode is a sequence drawn from a designed whitelist. Molecules synthesized in one droplet receive the same barcode. After sequencing, reads with the same valid cell barcode are assigned to the same putative cell.

A barcode is an assay-level identifier whose uniqueness is scoped to a library or run. Its uniqueness is scoped to the library or run. A complete cell key commonly includes sample or library identity plus the cell barcode. Concatenating those fields or using an explicit composite key prevents collisions across files.

6.4 Unique molecular identifiers

A UMI is a short random sequence attached to an original captured molecule. Multiple sequencing reads can originate from PCR copies of the same cDNA. Reads sharing cell barcode, gene assignment, and UMI are collapsed to estimate one original molecule. This reduces amplification bias.

UMIs are local molecular tags. Collisions can occur when two original molecules receive the same UMI within the same cell and gene context. Error-correction procedures also merge nearby UMI sequences under defined rules. The resulting matrix value is therefore an estimated molecule count under the pipeline's counting model.

6.5 Read alignment and gene assignment

The cDNA-derived read is aligned or pseudoaligned to a reference genome or transcriptome. The pipeline uses an annotation to assign compatible reads to genes. Reads overlapping exons, splice junctions, introns, or multiple loci can be treated differently depending on the software and configuration.

The matrix gene axis inherits the identifiers used by the reference bundle. A Cell Ranger reference, for example, is constructed from a FASTA assembly and GTF annotation. The feature table can use Ensembl gene IDs as primary feature IDs and gene symbols as display names. The resulting matrix therefore reflects both the biological sample and the reference annotation.

6.6 Counts, normalization, and transformed expression

Raw UMI counts are nonnegative integers. Downstream analysis can normalize each cell by library size, scale to a target total, apply $\log_{1p}$, select variable genes, or construct residual representations. The identifier axis remains conceptually independent of the numerical transformation.

The MANTRA raw atlas contract records `.x` as $\log_{1p}$ CP10K expression for the pinned slim atlas. This semantic statement is part of the file's data contract. A matrix with the same shape but raw counts would be a different computational object even if its `obs` and `var` identifiers were identical.

6.7 Pseudobulk aggregation

A pseudobulk profile aggregates single-cell expression values over a group, often all cells sharing a perturbation. For raw counts, aggregation can sum molecules before normalization. For transformed values, the chosen averaging and baseline procedure determine the resulting statistic. The perturbation identity defines the grouping axis; the gene identifiers define the vector coordinates.

Pseudobulk aggregation retains cell-level provenance as part of the aggregate definition. Quality-control masks, control matching, guide assignments, cell states, and batch variables determine which cells contribute to each aggregate.

6.8 AnnData and H5AD

AnnData is an annotated matrix data structure widely used in Python single-cell analysis. Its central matrix `X` has observations on rows and variables on columns. Observation metadata are stored in `obs`; variable metadata are stored in `var`. Additional matrices and metadata occupy `layers`, `obsm`, `varm`, `obsp`, `varp`, and `uns`. The `.h5ad` file format stores this structure in HDF5 [12].

AnnData component	Typical role in a single-cell perturbation atlas
<code>X</code>	cell-by-expression-gene matrix
<code>obs</code>	perturbation labels, guide identities, cell-level QC, batch, control status
<code>var</code>	gene symbols, Ensembl gene IDs, feature metadata, response-axis flags
<code>layers</code>	alternate count or normalized matrices with the same axes
<code>obsm</code>	cell embeddings, reduced coordinates, or other row-aligned matrices
<code>varm</code>	gene loadings or other column-aligned matrices
<code>uns</code>	unstructured metadata, parameters, palettes, or provenance

The HDF5 path supplies storage location, while the project manifest supplies the semantic contract. `obs/gene` means whatever the producing dataset defines. `var/_index` can be a symbol, Ensembl ID, or another key. The project manifest records the meaning, dtype, cardinality, and uniqueness expectations for required fields.

6.9 The two dimensions of an expression matrix

An expression matrix has a row identity and a column identity. In the atlas, rows are cells. Columns are expression-gene features. A perturbation label is row metadata rather than a column key. The same gene name can therefore appear in two roles:

- as the intended perturbation attached to a cell; and
- as one coordinate in the expression vector measured for that cell.

These roles are causally connected but structurally independent. A cell perturbed at gene A has measurements for gene A, gene B, gene C, and thousands of other response genes. The perturbation axis and expression axis can have different identifier systems, cardinalities, and duplicate-resolution rules.

6.10 The raw K562 atlas as an annotated matrix

The pinned raw atlas has shape [310385, 8563]: 310,385 cell observations and 8,563 expression-gene variables. Its SHA-256 is

```
51d6c7ecb349d34ec4c0e1bfb44a6d832e4113b26f46a319fc7993c028a5f060
```

The required cell metadata include perturbation identity, control status, quality-control masks, count metrics, and guide information. The required gene metadata include the atlas gene key, symbol, Ensembl gene ID, and project selection flags.

The file's identity combines exact bytes with semantic meaning. The hash establishes byte identity. The manifest establishes that `.X` means `log1p CP10K`, that `qc_pass_minimal` is the default cell mask, and that `var/in_gwps` and related fields have defined roles.

6.11 Direct guide capture as a second feature space

A direct-capture Perturb-seq experiment can produce a gene-expression library and a guide-capture library from the same cells. Both libraries contain cell barcodes. The guide reads are counted against a guide feature reference, while the cDNA reads are counted against a gene annotation. The two feature spaces are joined through cell barcodes.

The guide feature ID is therefore parallel to a gene feature ID at the assay level, but the biological object types differ. One is an experimental construct sequence. The other is a gene or transcript annotation feature. A multimodal container can store both, or a preprocessing pipeline can materialize the called guide identities in `obs`.

6.12 Controls and matched-control baselines

Non-targeting cells provide empirical control distributions. MANTRA additionally uses matched-control procedures to estimate baseline expression for treated cells. The control cell identity, matching procedure, and residual target representation are downstream modeling objects. They remain keyed by the same expression-gene response axis.

6.13 Data provenance from molecule to model

The complete provenance chain is:

cell and guide construct → RNA molecules → barcoded cDNA → sequencing reads → aligned features → UMI matrix

Every arrow applies an algorithm or experimental transformation. The identifier contract records enough context to reproduce the node and interpret the transformation.

7 MANTRA Identifier Architecture

7.1 Architectural goal

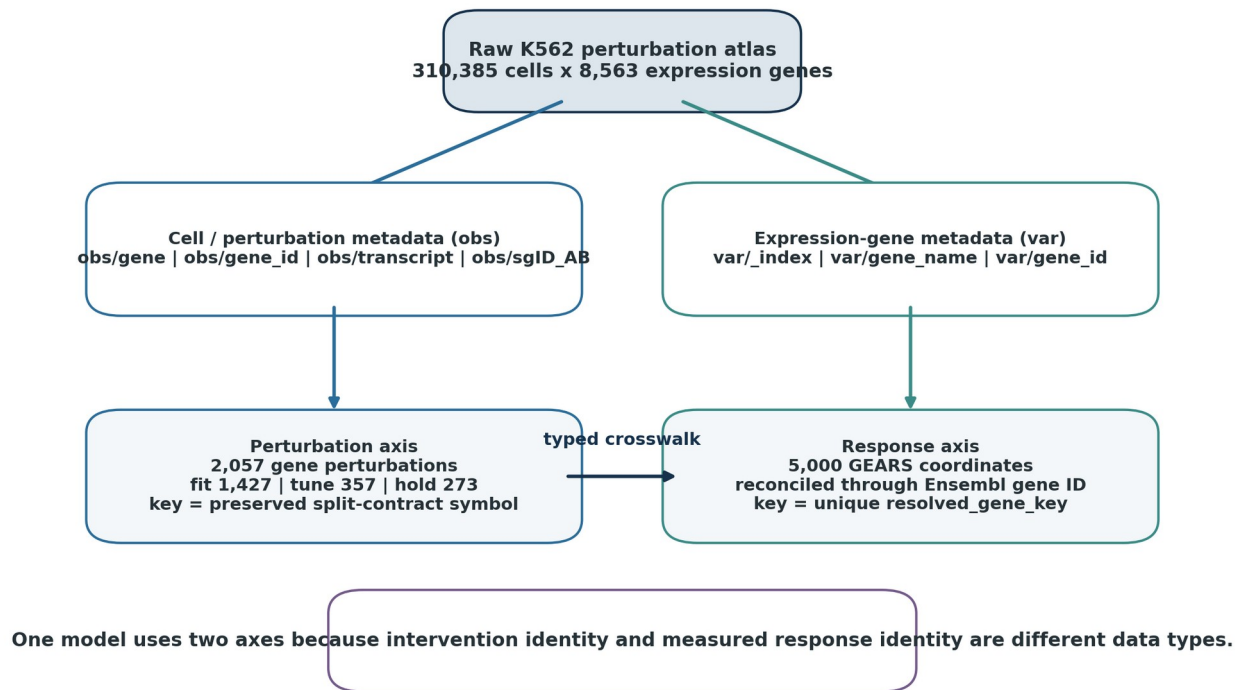
The MANTRA identity layer supports a perturbation-response model built from historical experimental labels, contemporary biological priors, and a separately reconciled response-gene surface. Its goal is to preserve every relevant identity at the level where it is valid while exposing stable ordered axes to model code.

The architecture has two canonical model-facing spaces:

1. the perturbation axis, whose rows are the exact gene-symbol labels inherited from the fit, tune, and hold split contract; and
2. the response-gene axis, whose columns are unique `resolved_gene_key` values constructed by reconciling the GEARS surface to atlas genes through Ensembl gene IDs.

These spaces are connected through typed metadata. They remain separate because the intervention row and predicted gene-expression coordinate are different objects.

MANTRA separates perturbation identity from response-gene coordinates



The MANTRA pipeline materializes separate perturbation and response axes.

7.2 Canonical object sets and notation

The raw and model-facing sets are:

Symbol	Meaning	Cardinality
<i>C</i>	atlas cell index set	310,385
<i>P</i>	raw perturbation-label set including non-targeting	2,058

Symbol	Meaning	Cardinality
	control	
P_{gene}	gene-target perturbation set	2,057
V	raw atlas expression-gene axis	8,563
Y	canonical GEARS response-gene axis	5,000
E	Ensembl gene-ID strings observed in the atlas	source-defined
T	transcript IDs or library transcript labels	source-defined
Q	paired-guide identities	source-defined

For cell $i \in C$, the raw experimental identity record is

$$R(i) = (p_i, e_i, t_i, q_i) \in P \times E \times T \times Q,$$

where p_i is the perturbation label, e_i is the raw Ensembl gene ID, t_i is the transcript identifier or library label, and q_i is the paired-guide identity.

The four fields are complementary. The perturbation label indexes the benchmark row. The Ensembl gene ID identifies the targeted locus in the source metadata. The transcript field preserves the promoter or transcript context of library design. The guide pair identifies the exact intervention construct.

7.3 Raw cell-level fields

The atlas cell metadata contain the following fields.

H5AD field	Object represented	Cardinality in source	ALG1L example
obs/gene	raw perturbation gene-symbol label	2,058 including control	ALG1L
obs/perturbation_id	perturbation label	2,058 including control	ALG1L
obs/gene_id	raw Ensembl gene ID	2,058 including control	ENSG00000189366
obs/transcript	target transcript ID or library transcript label	58 distinct values in the source summary	ENST00000340333.3
obs/gene_transcript	composite library label containing numeric prefix, symbol, transcript, and gene ID	2,273 distinct values	267_ALG1L_ENST00000340333.3_ENSG00000189366
obs/sgID_AB	exact paired-guide identity	2,273 distinct values	two ALG1L guide components

The non-targeting control occupies the perturbation-oriented fields with the literal experimental label non-targeting. The control row is represented through its experimental object type and control flag.

The composite `gene_transcript` and `paired-guide` fields show that the library had a more granular construct vocabulary than the gene-level benchmark axis. Multiple constructs can map to one perturbation symbol. The benchmark split intentionally indexes the gene-level intervention while the raw cell records retain construct-level provenance.

7.4 Perturbation split contract

The canonical GEARS split partitions the 2,057 gene perturbations into three ordered sets.

Split	Number of perturbation		Row identifier
	rows		
fit	1,427		exact raw perturbation gene symbol
tune	357		exact raw perturbation gene symbol
hold	273		exact raw perturbation gene symbol

The row strings include labels such as `FNTB`, `RBM14`, and `ALG1L`. The order is part of the contract. Model outputs, target arrays, priors, and evaluation files are aligned to these rows.

The perturbation identity is defined as

```
perturbation_identity := exact symbol string in split_contract.json
```

Current HGNC, Ensembl, NCBI, transcript, and protein identities are attached as fields. This preserves compatibility with source arrays and benchmark partitions while enabling modern biological interpretation.

7.5 Why the perturbation label remains historical

The guide library was designed under an annotation and nomenclature context available at design time. The label records the intervention as constructed and assayed. A later symbol update leaves the row's cells, guides, and stored arrays anchored to the original experimental identity. The current record is added as a resolution layer.

This is analogous to preserving an original foreign key from a source transaction while attaching a dimension-table key for current analytics. The source identity is immutable provenance. The resolved identity is a curated interpretation. Both are required for a stable warehouse.

7.6 Expression-gene metadata

The raw H5AD expression columns use three important identity fields.

H5AD field	Count	Role
<code>var/_index</code>	8,563 unique keys	exact H5AD column access
<code>var/gene_name</code>	8,561 unique symbols	human-readable display label
<code>var/gene_id</code>	8,563 Ensembl gene IDs	cross-resource locus reconciliation

The two-symbol difference reveals duplicate values in the display-symbol vector. The unique H5AD index and Ensembl gene IDs preserve the 8,563 columns.

7.7 Construction of the 5,000-gene response axis

The GEARS evaluation surface selects 5,000 response coordinates. The gene-axis table records, for each coordinate:

- the raw GEARS symbol;
- the GEARS Ensembl gene ID;
- the matched atlas symbol;
- the matched atlas Ensembl gene ID;
- the atlas unique column key; and
- the final `resolved_gene_key`.

Ensembl gene IDs supply the principal cross-resource reconciliation. The resolved key supplies uniqueness and stable model access. The display symbols remain available for reporting.

Formally, let $Y = \{0, \dots, 4999\}$ be the ordered response coordinates. The axis builder emits a vector

$$k_Y: Y \rightarrow K,$$

where K is the set of unique resolved key strings. It also emits metadata projections such as symbol $s_Y(y)$ and Ensembl ID $e_Y(y)$. The uniqueness invariant applies to k_Y , while s_Y can contain repeated values.

7.8 TBCE: duplicate display symbol, distinct response coordinates

The raw symbol TBCE appears at two response coordinates. Ensembl reconciliation distinguishes the loci.

Axis index	Raw symbol	Ensembl gene ID	Resolved gene key
535	TBCE	ENSG00000285053	TBCE
536	TBCE	ENSG00000284770	TBCE-1

A display symbol can label more than one response coordinate



The unique model key preserves both coordinates; the shared symbol remains available as a display label.

The response-axis key preserves both TBCE coordinates.

The suffix in TBCE-1 is a project-local uniqueness device. Its namespace is MANTRA and its object type is a response-coordinate key, distinct from Ensembl feature versions, HGNC symbols, protein isoforms, and RefSeq versions. Its namespace and object type identify it as a MANTRA response-coordinate key.

7.9 The two axes as distinct types

The perturbation and response axes can be written as separate types:

```
PerturbationRowKey = source experimental symbol in split order
ResponseColumnKey = unique resolved_gene_key in GEARS order
```

Each axis type defines its own valid values and metadata. A model prediction matrix with shape [273, 5000] has hold perturbation rows and response-gene columns. Its row metadata and column metadata are separate records.

This distinction becomes especially important when a perturbed gene is absent from the 5,000-gene response surface, when a response symbol is duplicated, or when historical perturbation nomenclature differs from current response-gene nomenclature.

7.10 Identifier flow

The identity flow can be represented as two branches from the raw atlas.

7.10.1 Perturbation branch

```
raw cell metadata
-> exact perturbation symbol
-> fit / tune / hold row
-> pinned HGNC resolution
-> Ensembl and NCBI cross-references
-> locus-type-dependent priors
-> model perturbation row
```

7.10.2 Response branch

```
raw atlas var metadata + GEARS axis
-> Ensembl gene-ID reconciliation
-> atlas unique column key
-> 5,000 unique resolved_gene_key values
-> model response columns
```

A biological crosswalk can connect a perturbation record to a response coordinate. The crosswalk records the mapping namespace and source. The axes remain separately ordered.

7.11 Required resolved identity record

A source-specific prior row should carry the following fields when applicable.

Field	Semantics
perturbation_identity	exact source row label
match_kind	approved symbol, previous symbol, alias, direct Ensembl match, control, or explicit resolution class
hgnc_id	stable HGNC nomenclature record
approved_symbol	approved HGNC symbol in the pinned snapshot
ensembl_gene_id	resolved Ensembl gene accession

Field	Semantics
ncbi_gene_id	NCBI GeneID as a string
locus_type	pinned nomenclature or annotation class
raw_transcript_id	transcript label carried by the experiment
current_transcript_ids	release-specific transcript records as needed
uniprot_accessions	validated protein entries or an explicit empty list
gencode_release	annotation release
ensembl_release	paired Ensembl database release
genome_assembly	coordinate assembly
source_snapshot_sha256	exact resolver or annotation bytes
resolution_provenance	resolver name, version, parameters, and timestamp or run ID

The record can be serialized as JSON, Parquet, TSV, or another typed format. Null and empty-list semantics should be defined. An empty protein list means that resolution completed and found no supported current protein accession under the project rule. A null field can mean that the field was outside the scope of the source.

7.12 Resolution precedence

The nomenclature resolver applies a deterministic precedence:

1. exact approved HGNC symbol match;
2. unique previous-symbol match;
3. unique alias-symbol match;
4. direct stable-ID cross-reference when the source provides one;
5. explicit ambiguous or unresolved state.

The procedure preserves the input string and returns the full candidate evidence. A direct source Ensembl gene ID can support or disambiguate symbol resolution. Transcript membership is evaluated inside the pinned annotation release.

7.13 Using Ensembl IDs for locus and matrix reconciliation

Ensembl gene IDs serve two related roles. In raw perturbation metadata, `obs/gene_id` describes the targeted locus as recorded by the atlas. In expression metadata, `var/gene_id` describes each measured gene feature. In the GEARS surface builder, Ensembl IDs reconcile source rows to atlas columns.

The shared namespace enables a join, while field role remains explicit. An Ensembl gene ID in `obs` is target metadata. An Ensembl gene ID in `var` is measurement-axis metadata. A typed schema can use names such as `target_ensembl_gene_id` and `response_ensembl_gene_id` when the fields coexist in one record.

7.14 Transcript interpretation

Raw transcript labels are preserved exactly because they document library design. Current transcript membership is computed against GENCODE release 44. The resolver distinguishes the stable transcript stem from the version and looks up the full accession when available.

A transcript stem can be reassigned or have a different version in a later release. The project records:

- raw transcript string;
- parsed namespace and stable stem;
- raw feature version;
- current release lookup result;
- current gene membership;
- match relation such as exact accession.version, stem-only continuation, historical context, or absent in release.

This representation makes release drift visible without rewriting the experiment.

7.15 Protein-derived prior coverage

Protein-derived priors require a locus that supports a current protein product and a resolved protein accession. The identity row records the chosen UniProt accession and sequence hash. The selection policy can prioritize reviewed entries, canonical sequences, or a defined isoform set.

For a pseudogene such as current ALG1L1P, an empty current UniProt accession list is a resolved outcome. Other source databases may contain historical or similarity-based protein references; the MANTRA record follows the pinned current-locus policy and retains the provenance of any alternate evidence separately.

7.16 Coordinate-derived prior coverage

Chromatin, regulatory, and genomic-neighborhood priors are computed from coordinate-bearing sources. Each row records:

- source assay or file accession;
- source assembly;
- annotation release used to define gene intervals;
- aggregation window and strand policy;
- transformation or lift-over metadata;
- exact file hashes; and
- output feature semantics.

This separates measured signal from the gene coordinates used to summarize it. An ENCODE file and a GENCODE GTF remain independent inputs joined through GRCh38.p14 coordinates.

7.17 Project-local identifiers

Project-local identifiers are legitimate namespaces when their domains are explicit. MANTRA defines at least two:

Local namespace	Domain	Example
perturbation identity	ordered experimental intervention rows	ALG1L
resolved gene key	ordered response-gene coordinates	TBCE - 1

A local key can remain stable even when external databases update, because the project contract pins the source data and mapping. External current identities can be refreshed in a new resolver snapshot while preserving the local row and column order.

7.18 Source artifact identity

Each source artifact has mechanistic and semantic identity.

Mechanistic identity includes:

- path or content-addressed location;
- SHA-256;
- byte count;
- array shape;
- dtype;
- required keys and fields; and
- producing script and command.

Semantic identity includes:

- what `.X` represents;
- which cell mask is the default;
- what each identifier field means;
- which axis order is canonical;
- whether the file contains held-out targets or labels; and
- what downstream uses are permitted.

The build report records how the artifact was produced. The manifest records what the artifact means and what a consumer can rely on.

8 Worked Identity Case Studies

8.1 Case study 1: ALG1L across experimental and current contexts

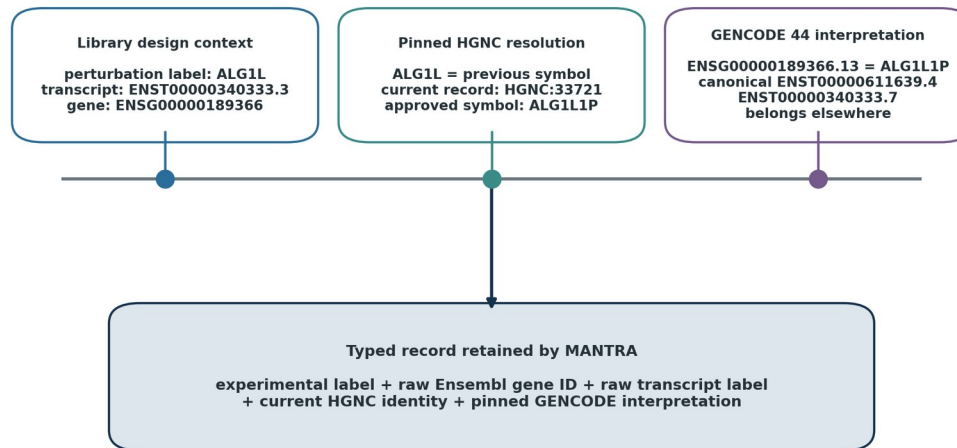
The raw atlas stores the following target metadata:

```
perturbation label: ALG1L
gene ID:           ENSG00000189366
transcript:        ENST00000340333.3
```

The pinned HGNC snapshot resolves ALG1L as a previous symbol for ALG1L1P, record HGNC:33721, and cross-references Ensembl gene ENSG00000189366. GENCODE release 44 classifies ENSG00000189366.13 as the transcribed pseudogene ALG1L1P and provides canonical transcript ENST00000611639.4.

GENCODE release 44 assigns transcript stem ENST00000340333, at version .7, to ENSG00000291096.1, a neighboring lncRNA record. The atlas's pair of ENSG00000189366 with ENST00000340333.3 therefore reflects an earlier annotation and library-design context.

ALG1L: one experimental label across historical and current annotation contexts



The record gains context over time while preserving the identity used by the experiment.

ALG1L gains current context while preserving its experimental identity.

The resulting record is layered rather than overwritten:

Layer	Identity
benchmark perturbation row	ALG1L
raw target Ensembl gene	ENSG00000189366
raw transcript label	ENST00000340333.3
current HGNC record	HGNC:33721
current approved symbol	ALG1L1P
GENCODE 44 gene feature	ENSG00000189366.13
GENCODE 44 representative transcript	ENST00000611639.4
current protein coverage	empty under the project's current protein-coding

Layer	Identity criterion
-------	-----------------------

This case demonstrates why a release-aware typed record is more informative than a replacement symbol. The experiment, current gene nomenclature, current annotation, and protein-coverage status are all available for their respective uses.

8.2 Case study 2: TBCE and response-axis uniqueness

The GEARS response surface contains two rows displayed as TBCE. Their Ensembl gene IDs are different. The axis builder preserves both and assigns unique resolved keys TBCE and TBCE-1.

The model can therefore predict both coordinates independently. Human-readable reports can show the shared raw symbol alongside Ensembl IDs. Array access uses the unique resolved key or integer position.

This is a general pattern. Display labels can collide because of historical source construction, symbol reuse, annotation choices, or multiple feature records. A matrix requires unique coordinate keys. The uniqueness layer should be explicit and local, while the source label and external accessions remain available.

8.3 Case study 3: TP53 across gene, transcript, sequence, and protein namespaces

A familiar protein-coding gene illustrates the ordinary one-to-many relations.

Namespace	Example object for the TP53 case
HGNC	gene nomenclature record HGNC:11998, approved symbol TP53
Ensembl gene	ENSG00000141510 gene feature lineage
Ensembl transcript	one of several ENST transcript features
RefSeq RNA	accessioned RNA sequence such as NM_000546.6
RefSeq protein	accessioned protein sequence such as NP_000537.3
UniProt	reviewed protein entry P04637, with represented isoforms
GENCODE	release-specific transcript structures and tags on GRCh38
ENCODE	experiments measuring chromatin, transcription-factor binding, or RNA at the locus

The row has several task-specific authoritative identifiers. HGNC is appropriate for approved gene nomenclature. Ensembl and GENCODE are appropriate for genomic feature structure. RefSeq is appropriate for a specific reference sequence. UniProt is appropriate for protein knowledge. An experiment can cite several simultaneously.

8.4 Case study 4: a long noncoding RNA locus

Consider an annotated lncRNA with an HGNC record, an Ensembl gene ID, several Ensembl transcript models, and one or more RefSeq noncoding RNA accessions. Its identity graph differs from a protein-

coding gene in one important branch: the ordinary gene → coding transcript → protein mapping can be empty.

The locus still has genomic coordinates, promoters, exons, splice junctions, expression measurements, chromatin context, and perturbation effects. CRISPRi can target its promoter. RNA-seq can quantify its transcripts. ENCODE can provide chromatin data. A protein-derived prior can remain absent while other priors are present.

This case shows gene identity and protein identity as separate projections linked by explicit gene-to-transcript-to-protein relations. Protein is one optional projection of a gene and transcript record.

8.5 Case study 5: a processed pseudogene

A processed pseudogene can resemble the exon-joined sequence of its parent transcript because it arose through RNA reverse transcription and genomic reinsertion. It can have an Ensembl gene ID, HGNC pseudogene symbol, genomic coordinate, and transcribed pseudogene annotation. It often lacks the parent gene's introns.

Short RNA-seq reads can map ambiguously between the parent gene and pseudogene. The feature-counting pipeline's annotation and multimapping policy determine whether the pseudogene receives counts. A protein sequence associated with the parent gene supplies protein identity for that parent record; pseudogene protein coverage is established from the pseudogene's own current record and evidence. Parent relation, locus identity, transcription, and protein coverage occupy separate fields.

8.6 Case study 6: one dual-guide perturbation record

A same-gene dual-guide CRISPRi element can be represented as:

```
perturbation_identity: FNTB
construct_type: same_gene_dual_crispri
guide_A_id: FNTB_guide_A
guide_B_id: FNTB_guide_B
guide_A_target: chr14:...:strand
guide_B_target: chr14:...:strand
raw_transcript_label: ENST...
target_ensembl_gene_id: ENSG...
effector: dCas9-repressor
library_element_id: sgID_AB string
```

The model row can aggregate all valid cells assigned to the FNTB perturbation. Construct-aware analyses can stratify by exact pair. Guide-level quality diagnostics can examine A and B separately. All three levels derive from the same raw record.

8.7 Case study 7: one cell from physical experiment to model tensor

An end-to-end cell record can be traced as follows:

1. A lentiviral dual-guide element is delivered to a K562 CRISPRi cell.
2. The guide pair represses transcription at its intended target promoter.
3. The cell contains a perturbed RNA state.
4. Polyadenylated RNAs are captured and reverse-transcribed into barcoded cDNA.
5. Sequencing reads recover cell barcode, UMI, transcript-derived insert, and guide-derived sequences.
6. The cDNA reads are assigned to annotation genes on the 8,563-column atlas axis.

7. The guide reads are called as a paired construct and materialized in `obs/sgID_AB`.
8. The cell is assigned a gene-level perturbation symbol in `obs/gene`.
9. Quality-control and control-matching procedures select or transform the cell.
10. Aggregation contributes to a perturbation response vector on the 5,000-column resolved axis.

At no stage does one identifier replace all others. Each stage creates or carries an identity appropriate to its object.

8.8 Case study 8: ENCODE signal summarized to a gene prior

Suppose an ENCODE file contains a transcription-factor ChIP-seq signal track on GRCh38. The file has an ENCFE accession and belongs to an ENCSR experiment. To produce a gene-level feature, the pipeline selects promoter intervals from GENCODE release 44, summarizes signal around each promoter, and maps the resulting values to perturbation rows through HGNC or Ensembl gene identity.

The resulting prior row includes:

- ENCODE experiment and file accessions;
- downloaded file hash;
- source assembly;
- GENCODE release and GTF hash;
- promoter definition and window;
- aggregation statistic;
- target gene crosswalk; and
- final perturbation row order.

The ENCODE accession identifies measured data, the GENCODE feature identifies the interval, and the MANTRA perturbation label identifies the output row. Their orthogonality is the mechanism that makes the computation auditable.

9 Implementation and Governance

9.1 The identity layer as a typed subsystem

The implementation should expose biological identity as a typed subsystem rather than a collection of ad hoc string columns. The subsystem receives source records, parses namespace-specific syntax, resolves current and release-specific relations, emits ordered model axes, and records provenance.

A practical internal type hierarchy can include:

```

Identifier
├── GeneRecordIdentifier
│   ├── HgncId
│   ├── EnsemblGeneId
│   └── NcbiGeneId
├── TranscriptIdentifier
│   ├── EnsemblTranscriptId
│   └── RefSeqRnaAccession
├── ProteinIdentifier
│   ├── EnsemblProteinId
│   ├── RefSeqProteinAccession
│   └── UniProtAccession
├── AssayIdentifier
│   ├── EncodeExperimentAccession
│   └── EncodeFileAccession
├── ExperimentalIdentifier
│   ├── PerturbationIdentity
│   ├── GuideIdentity
│   └── GuidePairIdentity
└── ModelCoordinateIdentifier
    ├── PerturbationRowKey
    └── ResolvedGeneKey
  
```

The classes can be implemented as dataclasses, Pydantic models, Arrow schemas, or language-specific algebraic data types. The essential property is that a transcript accession cannot silently enter an API parameter requiring a gene-record identifier.

9.2 Parsing identifier syntax

Parsing separates syntax from biological resolution. An Ensembl parser can identify prefix, stable stem, and optional feature version. A RefSeq parser can identify prefix, accession body, and sequence version. A UniProt parser can separate primary accession and isoform suffix. An ENCODE parser can classify the record type from its prefix.

Example parsed objects:

```

{
  "namespace": "ensembl",
  "object_type": "transcript",
  "accession": "ENST00000611639",
  "version": 4
}
{
  "namespace": "refseq",
  "object_type": "rna_sequence",
  "accession": "NM_000546",
  "version": 6
}
  
```



```
{
  "namespace": "uniprot",
  "object_type": "protein_isoform",
  "accession": "P04150",
  "isoform": 2
}
```

Syntax parsing can be deterministic and release-independent. Resolution then consults pinned catalogs to determine existence, status, relations, and current attributes.

9.3 Resolution result as a first-class object

A resolver should return a structured result rather than only the chosen symbol. One useful schema is:

```
input_value: ALG1L
input_field: perturbation_identity
input_namespace: experimental_symbol
status: resolved
match_kind: previous_symbol
selected:
  hgnc_id: HGNC:33721
  approved_symbol: ALG1L1P
  ensembl_gene_id: ENSG00000189366
candidates:
  - hgnc_id: HGNC:33721
    evidence:
      - previous_symbol_exact
      - source_ensembl_cross_reference_agreement
resolver:
  catalog_sha256: ...
  code_sha256: ...
  policy_version: hgnc_resolution.v1
```

This record preserves the decision and the evidence. Ambiguity is represented by several candidates and a status, rather than by an arbitrary selected row. The source Ensembl ID can be incorporated as independent evidence.

9.4 Crosswalk tables

A crosswalk table materializes release-specific relations among namespaces. Each row can include:

Field	Purpose
source namespace and accession	left endpoint
target namespace and accession	right endpoint
source and target object types	semantic typing
relation type	exact, parent, transcript-of, protein-product-of, previous-symbol-of, alias-of, MANE match, or other relation
source release	state of the left database
target release	state of the right database
evidence source	file, API, GTF, or curated catalog
confidence or status	exact, unique, multiple, deprecated, or unresolved
provenance hash	content identity of the crosswalk source

A crosswalk can contain multiple rows for one source accession. Consumers then select relations appropriate to their task. A protein prior can request current reviewed protein products. A transcript analysis can request every transcript in the pinned annotation. A display layer can request the current approved symbol.

9.5 Ordered axis contracts

An axis contract defines both membership and order. A perturbation-axis contract can be represented as:

```
{
  "axis_id": "gears_5k_perturbations.v1",
  "object_type": "experimental_gene_perturbation",
  "key_field": "perturbation_identity",
  "splits": {
    "fit": 1427,
    "tune": 357,
    "hold": 273
  },
  "source_sha256": "...",
  "ordered_keys_sha256": "..."
}
```

A response-axis contract similarly records 5,000 ordered resolved keys, their Ensembl IDs, source rows, and a hash of the ordered vector. Hashing the serialized ordered key vector provides a compact check that two arrays share the same column semantics.

9.6 Invariants for the K562 atlas contract

The following invariants define the pinned identity surface:

1. The raw H5AD has SHA-256 51d6c7ecb349d34ec4c0e1bfb44a6d832e4113b26f46a319fc7993c028a5f060.
2. The matrix has 310,385 cell rows and 8,563 expression-gene columns.
3. Cell metadata represent 2,057 gene-target perturbations and one non-targeting control label.
4. The split contract contains 1,427 fit, 357 tune, and 273 hold perturbation rows.
5. The perturbation key vector preserves exact split-contract strings and order.
6. Approved, previous-symbol, and alias resolutions record a unique selected HGNC record or an explicit candidate state.
7. Protein-derived rows record the chosen protein accession and exact sequence identity.
8. Coordinate-derived rows record assembly, annotation source, and source-file hash.
9. The 5,000 response keys are unique.
10. Ensembl reconciliation retains both TBCE coordinates.

These invariants can be evaluated in unit tests and artifact-registration scripts. They transform the identity layer from descriptive documentation into an executable contract.

9.7 Build reports and manifests

MANTRA separates build provenance from read semantics.

A build report answers:

- Which script wrote the artifact?
- Which command and parameters were used?
- Which input bytes were read?

- What output bytes, shape, dtype, and fields were produced?
- Which source-control commit and resolved configuration were active?

A manifest answers:

- What does the artifact represent?
- Which fields and masks define its canonical views?
- Which identifier system and object type does each axis use?
- Which downstream operations are supported?
- Which benchmark or legality constraints apply?

The build report is mechanistic provenance. The manifest is a semantic interface. A consumer validates both before reading an artifact.

9.8 Suggested identity schema

A normalized project table can use one row per perturbation identity with nested or linked tables for one-to-many relations.

9.8.1 Core perturbation table

Field	Type	Meaning
perturbation_identity	string, primary key	exact split row label
split	enum	fit, tune, or hold
split_index	integer	ordered position within split
hgnc_id	nullable string	selected nomenclature record
approved_symbol	nullable string	approved symbol in pinned snapshot
match_kind	enum	resolution relation
ensembl_gene_id	nullable string	selected gene feature stem
ncbi_gene_id	nullable string	NCBI GeneID
locus_type	nullable string	source locus class
resolution_status	enum	resolved, ambiguous, unresolved, control
resolver_provenance_id	foreign key	resolver run and source snapshot

9.8.2 Raw experimental mapping table

Field	Type	Meaning
cell_or_construct_key	string	source record key
perturbation_identity	foreign key	benchmark row
raw_gene_id	string	source Ensembl gene ID or control label
raw_transcript	string	source transcript or library label
guide_A_id	string	first guide identity

Field	Type	Meaning
guide_B_id	string	second guide identity
guide_pair_id	string	exact sgID_AB
source_file_sha256	string	raw atlas identity

9.8.3 Protein mapping table

Field	Type	Meaning
perturbation_identity	foreign key	project row
uniprot_accession	string	protein entry
isoform	nullable string	selected isoform
review_status	enum	reviewed or unreviewed
sequence_sha256	string	exact amino-acid sequence bytes
mapping_relation	string	cross-reference path
source_release	string	UniProt release or retrieval snapshot

9.8.4 Response-axis table

Field	Type	Meaning
axis_index	integer, primary key	integer tensor coordinate
resolved_gene_key	unique string	model-facing coordinate key
raw_gears_symbol	string	source display label
gears_ensembl_gene_id	string	source locus identity
atlas_var_key	string	exact raw atlas column key
atlas_gene_symbol	string	atlas display label
atlas_ensembl_gene_id	string	reconciled atlas locus identity
mapping_status	enum	exact, reconciled, or source- defined status

9.9 API design

A concrete API can separate parsing, resolution, and axis construction.

```

parsed = parse_identifier("ENST00000611639.4")
# ParsedIdentifier(namespace="ensembl", object_type="transcript", ...)

resolution = hgnc_catalog.resolve_symbol(
    symbol="ALG1L",
    supporting_ensembl_gene_id="ENSG00000189366",
)

record = build_perturbation_identity_record(
    perturbation_identity="ALG1L",
    raw_ensembl_gene_id="ENSG00000189366",
    raw_transcript="ENST00000340333.3",
    hgnc_resolution=resolution,
    gencode_catalog=gencode_44,

```

```
)

axis = build_response_axis(
    gears_rows=gears_gene_table,
    atlas_var=atlas_var_table,
    join_key="ensembl_gene_id",
)
```

Each operation returns a typed value with provenance. The axis builder emits diagnostics and the ordered key hash. The resolver exposes candidates and match evidence.

9.10 Diagnostic summaries

Useful diagnostics include:

- counts by resolution status and match kind;
- approved versus previous-symbol versus alias matches;
- one-to-many transcript and protein relation counts;
- locus-type distribution;
- protein-coverage rate by locus type;
- exact versus stem-only transcript-version matches;
- response-axis duplicate symbol counts;
- guide-pair multiplicity per perturbation;
- cells per construct and perturbation;
- source-release and assembly coverage; and
- hashes of every ordered axis.

Diagnostics are most useful when they are emitted as structured tables as well as human-readable reports. The structured form supports regression tests across rebuilds.

9.11 Versioned policy

Resolution is partly a policy. Examples include the precedence among approved, previous, and alias symbols; the choice of UniProt reviewed versus unreviewed entries; the handling of transcript stems whose current membership changed; and the construction of unique response keys.

The policy should have a versioned identifier, such as `identifier_resolution.v1`. A policy update creates a new resolver output and comparison report. The source experimental axis can remain fixed while current metadata are refreshed under a new policy.

9.12 Comparison between resolver versions

A resolver comparison should classify every changed row:

Change class	Example
nomenclature update	approved symbol changed while HGNC ID remained stable
cross-reference update	HGNC record gained or changed an Ensembl link
annotation update	transcript set or gene biotype changed
protein update	reviewed entry or sequence changed

Change class	Example
ambiguity change	alias became unique or newly ambiguous
local-axis invariant	perturbation and response keys remained byte-identical

The report allows biological metadata to evolve without obscuring model-coordinate stability.

9.13 Provenance hashes and content addressing

Cryptographic hashes identify exact file bytes. For compressed files, the hash of the distributed compressed bytes and the hash of decompressed semantic content can both be useful. Sequence caches can hash canonical FASTA bytes after a documented normalization. Ordered vectors can hash newline-delimited UTF-8 keys.

A hash establishes whether two artifacts are byte-identical under the chosen serialization. The manifest describes semantics. The manifest supplies semantics, and tests connect the two.

9.14 Human-readable documentation and machine-readable contracts

The monograph explains why the architecture exists. Machine-readable schemas enforce it. Every important table in this document should have a corresponding contract in JSON Schema, Pydantic, Arrow, YAML, or test code. The prose and schema should share field names and version identifiers.

The recommended documentation stack has three layers:

1. this conceptual monograph;
2. a concise engineering specification containing schemas, algorithms, and invariants; and
3. artifact-local manifests and build reports containing exact resolved values.

9.15 Canonical operational summary

For day-to-day development, the source of truth is:

Object	Canonical project identity
experimental perturbation row	symbol in <code>split_contract.json</code> , inherited from atlas obs/gene
raw cell-level target locus	atlas obs/gene_id
raw targeted transcript context	atlas obs/transcript
exact paired intervention	atlas obs/sgID_AB
current human nomenclature	pinned HGNC record
release-specific genomic structure	GENCODE 44 on GRCh38.p14 with Ensembl 110
protein identity	pinned UniProt accession, isoform policy, and sequence hash
functional-genomics assay identity	ENCODE record and exact file accession
raw expression column	atlas var/_index with var/gene_id metadata
model response coordinate	unique resolved_gene_key

The canonical implementation is a typed identity record that preserves each field at the biological, experimental, database, or computational level where it is defined.

10 Reference Appendices

Appendix A. Mental models

A.1 The biological production chain

```
reference assembly
  -> annotated gene locus
    -> primary RNA transcript
      -> RNA processing and splicing
        -> mature transcript isoform
          -> optional translated protein isoform
```

The arrows represent biological or annotation relations. Each node can have its own identifier. The gene-to-transcript and transcript-to-protein relations can be one-to-many or partial.

A.2 The experimental observation chain

```
guide construct
  -> perturbed cell
    -> cellular RNA molecules
      -> barcoded cDNA molecules
        -> sequencing reads
          -> gene and guide assignments
            -> H5AD cell-by-gene matrix
              -> MANTRA perturbation and response axes
```

The chain creates experimental and computational identities. A cell barcode and UMI are local assay identifiers. A perturbation identity and resolved response key are project-level identities.

A.3 The database projection chain

```
one biological case
  -> HGNC nomenclature record
  -> Ensembl gene and transcript features
  -> GENCODE release-specific structure
  -> NCBI Gene record
  -> RefSeq sequence accessions
  -> UniProt protein entry
  -> ENCODE assay records
  -> MANTRA local axis keys
```

The branches are parallel projections, each retaining its own namespace and object type.

Appendix B. Object-level reference table

Object level	Concrete object	Primary identifiers	Context fields
experimental perturbation	one intended intervention row	MANTRA perturbation identity	split, source atlas, guide construct
guide	one sgRNA sequence and vector position	guide ID	protospacer, scaffold, assembly, coordinate, position A/B
guide pair	one dual-guide library element	sgID_AB or construct ID	guide A, guide B, architecture, recombination state
gene nomenclature	one human nomenclature record	HGNC ID, approved symbol	snapshot, previous symbols, aliases, locus

Object level	Concrete object	Primary identifiers	Context fields
genome gene feature	one release-tracked locus feature	Ensembl ENSG	type feature version, release, assembly
transcript feature	one annotated RNA structure	Ensembl ENST	version, release, gene membership, exons
reference RNA sequence	one accessioned RNA sequence	RefSeq NM_, NR_, XM_, XR_	sequence version, release
protein feature	one translated annotation feature	Ensembl ENSP	version and release
protein knowledge entry	one UniProt entry or isoform	UniProt accession	isoform, review status, sequence hash, release
annotation package	one published gene set	GENCODE release	Ensembl release, assembly, source file hash
functional experiment	one assay	ENCODE ENCSR	biosample, target, method, replicates
functional data file	one downloadable file	ENCODE ENCF	file format, assembly, pipeline, hash
expression coordinate	one raw H5AD column	var/_index	symbol and Ensembl gene ID
response coordinate	one model output column	resolved_gene_key	axis index, symbol, Ensembl gene ID

Appendix C. Prefix and suffix reference

Pattern	Interpretation
HGNC:<number>	stable HGNC nomenclature record
ENSG...	Ensembl gene feature
ENST...	Ensembl transcript feature
ENSP...	Ensembl translated protein feature
ENSE...	Ensembl exon feature
.number after Ensembl feature	feature version under Ensembl rules
NM_...	RefSeq protein-coding RNA sequence
NR_...	RefSeq noncoding RNA sequence
NP_...	RefSeq protein sequence
XM..., XR..., XP...	predicted RefSeq RNA or protein sequence
.number after RefSeq accession	sequence version
UniProt base accession	protein knowledgebase entry
-number after UniProt accession	represented protein isoform
ENCSR...	ENCODE experiment record

Pattern	Interpretation
ENCFF...	ENCODE file record
GRCh38.p14	human assembly build 38, patch 14
project suffix such as TBCE-1	local unique coordinate key under the named project namespace

Appendix D. Guide-pair terminology

Term	Meaning in this document
sgRNA	single-guide RNA molecule or sequence design
guide A / guide B	positional components of a paired cassette
single-guide element	one construct carrying one sgRNA
same-gene dual guide	one construct carrying two sgRNAs directed to one gene or promoter context
combinatorial dual guide	one construct carrying sgRNAs directed to two different genes or loci
flanking guide pair	two nuclease guides defining the boundaries of a deletion
guide-pair ID	identifier for the complete A/B construct
direct guide capture	sequencing guide-derived molecules with cell-barcode linkage
guide assignment	called relation between a cell barcode and one or more guide identities
non-targeting control	experimental guide designed as a control under the library criteria
dual index	sequencing-library sample-index architecture; separate from dual-guide perturbation design

Appendix E. Glossary

Accession. A stable identifier assigned to a database record. An accession can have a version suffix when the namespace versions individual features or sequences.

Alias symbol. A symbol associated with an HGNC record through literature or database usage that is distinct from the current approved symbol and previous approved symbols.

Annotation. A structured interpretation of a genome assembly that defines gene, transcript, exon, coding-sequence, and other features.

Assembly. A named reference representation of a genome and its coordinate system.

Barcode. A designed sequence used to associate a sequencing molecule with a cell, sample, guide, or other experimental unit. Barcode scope is determined by the assay.

Biotype. A controlled category assigned to a gene or transcript feature, such as protein-coding, lncRNA, retained intron, or processed pseudogene.

Canonical transcript. A transcript selected as a representative under a particular resource's policy.

cDNA. DNA synthesized from an RNA template by reverse transcriptase. In RNA sequencing it is the amplifiable DNA representation of captured RNA.

Cell barcode. A sequence identifying the droplet or cell partition from which a set of molecules originated.

Chromatin. DNA packaged with histones and other proteins. Chromatin state influences regulatory accessibility.

Coding sequence (CDS). The portion of a transcript translated into a protein sequence.

Contig. A contiguous assembled sequence that can form part of a chromosome or remain unplaced.

Cross-reference. A database assertion connecting a record to a record in another namespace.

CRISPR activation (CRISPRa). Guide-directed recruitment of transcriptional activators to increase expression.

CRISPR interference (CRISPRi). Guide-directed recruitment of a catalytically inactive Cas effector and transcriptional repressor to reduce expression.

Display symbol. A human-readable gene label presented in tables and plots. It can have different uniqueness properties from a database accession or model key.

Dual-guide element. A library construct encoding two sgRNAs, usually represented as guide A and guide B.

Enhancer. A cis-regulatory genomic element that can increase transcription from a promoter in a context-dependent manner.

Exon. A segment retained in a mature transcript after RNA splicing. Exons can contain coding and untranslated sequence.

Feature. An annotated genomic object such as a gene, transcript, exon, or coding-sequence interval.

Feature version. A revision number attached to an individual Ensembl feature record according to object-specific rules.

Gene. A locus-level biological entity associated with one or more RNA products or other recognized functional outputs.

Gene model. A release-specific annotation representation of a gene locus and its associated transcript structures.

Gene symbol. A human-readable short label for a gene, such as an HGNC-approved symbol or a historical experimental symbol.

Genome. The complete genetic material of an organism or cell.

GTF. A tabular annotation format whose rows encode genomic features, coordinates, and attributes.

Guide RNA. An RNA molecule that directs a CRISPR effector to a sequence through base pairing and PAM-compatible targeting.

Guide-pair identity. The exact identifier of a construct containing guide A and guide B.

Haplotype. A linked set of sequence variants on one chromosome copy.

HGNC ID. Stable identifier for a human gene-nomenclature record.

Intron. A transcribed segment removed from a primary RNA during splicing.

Isoform. One of several transcript structures or protein sequences associated with a gene. The object type should be specified.

Lift-over. A coordinate transformation between genome assemblies based on sequence alignments or mapping chains.

lncRNA. Long noncoding RNA, an operationally defined heterogeneous class of noncoding transcripts commonly longer than 200 nucleotides.

Locus. A location or interval in the genome associated with a biological or annotation feature.

MANE. A cross-resource matched transcript set in which selected RefSeq and Ensembl/GENCODE transcript structures and sequences agree.

Manifest. A machine-readable semantic contract describing an artifact's meaning, fields, axes, and permitted uses.

Namespace. The issuing identification system and rules under which an accession has meaning.

NCBI GeneID. Stable numeric identifier for an NCBI Gene record.

Noncoding RNA. An RNA whose primary annotated product is the RNA rather than a conventional protein.

Object type. The class of entity identified, such as gene record, transcript feature, protein entry, assay file, guide pair, or model coordinate.

Ortholog. A gene in another species related through speciation from a gene in the common ancestor.

Paralog. A related gene produced through duplication within a lineage.

PAM. Protospacer-adjacent motif recognized by a Cas protein near the guide target site.

Perturbagen. The agent or intervention applied to alter a biological system.

Perturbation identity. The project key for an experimental intervention row. In MANTRA it is the exact split-contract symbol inherited from the atlas.

Poly(A) tail. A run of adenosines added to the 3' end of many mature eukaryotic RNAs and used for oligo-dT capture.

Primary transcript. The initial RNA synthesized from DNA before completion of processing such as splicing.

Promoter. A genomic region where transcriptional machinery assembles to initiate transcription.

Protein isoform. A distinct amino-acid sequence associated with the same gene or protein entry.

Pseudobulk. An aggregate expression profile constructed from a group of single cells.

Pseudogene. A gene-like locus with homology to a functional ancestral gene and a present annotation reflecting disrupted or altered protein-coding capacity.

Release. A published database-wide state containing a coherent set of records and relations.

Resolved gene key. A unique MANTRA response-axis key constructed for one model coordinate.

Reverse transcription. Synthesis of DNA from an RNA template.

Sequence version. A suffix identifying a revision of an accessioned sequence, as in RefSeq accession.version.

sgID_AB. The atlas field holding the exact paired guide A/B identity.

Splice junction. The connection between two exons in a processed transcript.

Transcript. A specific RNA structure produced from a locus, represented by an ordered set of exons and transcript boundaries.

Transcript isoform. One alternative mature RNA structure associated with a gene.

Transcript model. An annotation record representing the inferred structure and attributes of one transcript class.

UMI. Unique molecular identifier used to distinguish original captured molecules before PCR amplification.

UniProt accession. Identifier for a UniProt protein knowledgebase entry; an isoform suffix can identify a represented isoform.

Version. A revision identifier attached to an individual feature or sequence record under namespace-specific rules.

Appendix F. Evidence inventory for the MANTRA implementation

Evidence artifact	Relevant content
raw K562 perturbation atlas	obs/gene, obs/gene_id, obs/transcript, obs/gene_transcript, obs/sgID_AB, var/gene_name, var/gene_id
raw atlas manifest	matrix shape, .X semantics, required metadata, SHA-256
split contract	ordered fit, tune, and hold perturbation rows
response gene-axis table	raw symbols, Ensembl IDs, atlas keys, resolved keys
resolved-key vector	exact 5,000 model response-coordinate keys
HGNC snapshot	approved, previous, and alias symbols; stable IDs; cross-references; locus types
GENCODE source manifest	release, paired Ensembl release, assembly, URL, SHA-256
GENCODE GTF	gene, transcript, exon, CDS, coordinate, biotype, and HGNC attributes

Evidence artifact	Relevant content
UniProt FASTA cache	selected accessions and exact sequence bytes
identifier resolver	typed resolution result, match kind, candidates, and provenance
sequence coverage report	resolved gene-to-protein coverage and exact source identities
build reports	producing scripts, commands, input hashes, output hashes, and shapes

Appendix G. Selected references and authoritative resources

1. Bruford EA, Braschi B, Denny P, et al. “Guidelines for human gene nomenclature.” *Nature Genetics* 52, 754-758 (2020). DOI: 10.1038/s41588-020-0669-3.
2. HUGO Gene Nomenclature Committee. “Symbol report help” and HGNC downloadable field definitions. <https://www.genenames.org/> and <https://hgnc.genenames.org/help/symbol-report/index.html>
3. Ensembl. “Stable IDs: format and versioning.” https://mart.ensembl.org/info/genome/stable_ids/index.html
4. Frankish A, Diekhans M, Jungreis I, et al. “GENCODE: reference annotation for the human and mouse genomes in 2023.” *Nucleic Acids Research* 51, D942-D949 (2023). DOI: 10.1093/nar/gkac1071.
5. GENCODE. “The Human GENCODE Release History.” <https://www.encodegenes.org/human/releases.html>
6. GENCODE. “Gene and transcript biotypes.” <https://www.encodegenes.org/pages/biotypes.html>
7. UniProt Consortium. “UniProt: the Universal Protein Knowledgebase in 2025.” *Nucleic Acids Research* 53, D609-D617 (2025). DOI: 10.1093/nar/gkae1010.
8. UniProt. “UniProtKB help.” <https://www.uniprot.org/help/uniprotkb>
9. NCBI. “NCBI Gene overview and processing.” <https://www.ncbi.nlm.nih.gov/datasets/docs/v2/data-processing/gene-processing/gene/>
10. NCBI. “RefSeq Frequently Asked Questions” and “About RefSeq.” <https://www.ncbi.nlm.nih.gov/books/NBK50679/> and <https://www.ncbi.nlm.nih.gov/refseq/about/>
11. ENCODE Project. “Data organization” and “Getting started.” <https://www.encodeproject.org/help/data-organization/> and <https://www.encodeproject.org/help/getting-started/>
12. scverse. “AnnData” and “On-disk format.” <https://anndata.scverse.org/> and <https://anndata.readthedocs.io/en/stable/fileformat-prose.html>
13. Replogle JM, Saunders RA, Pogson AN, et al. “Mapping information-rich genotype-phenotype landscapes with genome-scale Perturb-seq.” *Cell* 185, 2559-2575.e28 (2022). DOI: 10.1016/j.cell.2022.05.013.
14. Replogle JM, Bonnar JL, Pogson AN, et al. “Maximizing CRISPRi efficacy and accessibility with dual-sgRNA libraries and optimal effectors.” *eLife* 11:e81856 (2023). DOI: 10.7554/eLife.81856.
15. Replogle JM, Norman TM, Xu A, et al. “Combinatorial single-cell CRISPR screens by direct guide RNA capture and targeted sequencing.” *Nature Biotechnology* 38, 954-961 (2020). DOI: 10.1038/s41587-020-0470-y.

16. 10x Genomics. “Getting Started: Single Cell 3’ Gene Expression.” Official workflow documentation. <https://www.10xgenomics.com/support/>
17. 10x Genomics. “Glossary of terms relevant to single-cell products,” entries for cell barcode and UMI. <https://www.10xgenomics.com/support/software/cell-ranger/>
18. Mattick JS, Amaral PP, Carninci P, et al. “Long non-coding RNAs: definitions, functions, challenges and recommendations.” *Nature Reviews Molecular Cell Biology* 24, 430-447 (2023). DOI: 10.1038/s41580-022-00566-8.
19. Pei B, Sisu C, Frankish A, et al. “The GENCODE pseudogene resource.” *Genome Biology* 13:R51 (2012). DOI: 10.1186/gb-2012-13-9-r51.
20. NCBI MeSH. “Reverse Transcription” and “Reverse Transcriptase Polymerase Chain Reaction.” <https://www.ncbi.nlm.nih.gov/mesh/68048348> and <https://www.ncbi.nlm.nih.gov/mesh/68020133>
21. NCBI. “What are accession numbers at NCBI?” NLM Customer Support Center. <https://support.nlm.nih.gov/knowledgebase/article/KA-03434/en-us>
22. GENCODE. “The GENCODE Project” and data-access documentation. <https://www.encodegenes.org/pages/genencode.html> and https://www.encodegenes.org/pages/data_access.html
23. HUGO Gene Nomenclature Committee. “HGNC Guidelines,” including pseudogene and RNA nomenclature. <https://hgnc.genenames.org/about/guidelines/>
24. MANTRA project-local evidence. GENE_IDENTIFIER_SYSTEMS_WHITE_PAPER.pdf, raw atlas manifests, split contract, response-gene axis table, GENCODE source manifest, and identifier-resolution reports pinned by the project.

Appendix H. Final canonical answer

A consistent gene-identification system for the MANTRA perturbation stack is a typed record with several orthogonal fields. The experimental perturbation row is the exact symbol inherited from the split contract. The raw atlas supplies target Ensembl gene IDs, transcript labels, and guide-pair identities. HGNC supplies current human gene nomenclature. Ensembl supplies stable gene and transcript feature records. GENCODE release 44 on GRCh38.p14 supplies release-specific genomic structure. NCBI Gene and RefSeq supply gene records and reference sequences. UniProt supplies protein knowledge and exact sequence identities. ENCODE supplies assay and file provenance. The 5,000-gene model response surface uses unique `resolved_gene_key` values reconciled through Ensembl gene IDs.

The complete semantics are preserved by a collection of typed projections and the versioned relations among them. This collection is the stable computational representation.